

When can non-veridical preferential attitude predicates take questions?

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Abstract

A growing body of evidence suggests that whether or not attitude predicates may combine with question complements is determined at least in part by some of their semantic properties. Zooming in, non-veridical preferential predicates (NVPs) such as *hope* and *fear* have been claimed not to combine with questions but this empirical generalization as well as its proposed explanation have been challenged by a number of counterexamples, whose properties—beyond the fact that they feature embedded questions—remain ill understood. By taking a closer look at the combinatorial and semantic properties of different NVPs from English, Japanese, Mandarin Chinese, Spanish and Turkish, this work narrows down the semantic properties of attitude predicates that determine their combinatorial properties in a cross-linguistically informed and predictive way. We propose a new classification of NVPs based on two empirical properties: C(lausal)-distributivity and valence, and show that non-C-distributive NVPs (e.g., *worry*) as well as evaluatively negative C-distributive NVPs (e.g., *fear*) can take questions in a canonical way (i.e., as their semantic arguments). For positive C-distributive NVPs such as *hope*, we suggest that the attested examples where *hope* appears to take questions involve non-canonical, adjunction-style composition.

Keywords: clause embedding, attitude predicates, selectional restrictions, veridicality, preferentiality, clausal distributivity

1 Introduction

One of the long-standing questions in the semantics of attitude predicates concerns whether the combinatorial (or selectional) restrictions of attitude predicates are related to their lexical semantic properties, and if so, how. In the literature, several works have engaged with a particular instance of this question, namely, the one pertaining to the distinction between so-called RESPONSIVE PREDICATES, i.e., predicates that can take both declarative and interrogative clauses, and ANTI-ROGATIVE PREDICATES, i.e., predicates that can take declarative but not interrogative clauses (Zuber, 1982; White and Rawlins, 2018; Theiler et al., 2019; Uegaki and Sudo, 2019; Özyıldız, 2021; Roberts, 2021; Djärv, 2023).¹ In (1) are examples of responsive (1a) and anti-rogative (1b) predicates:

- (1) a. Mary **knew/imagined/hated** {that Sue would call / who would call}.
b. Mary **believes/hopes** {that Sue will call / *who will call}.

These works observe correlations between the combinatorial restrictions of the relevant predicates and their semantic properties, such as veridicality, neg-raising, preferentiality and eventivity, and attempt to explain the restrictions based on these semantic properties. In particular, Theiler et al. (2019), Mayr (2019) and Uegaki and Sudo (2019) provide a concrete account of the combinatorial restrictions of (some classes of)

¹Following Dayal (2016), in this paper we use the terms *interrogative (clause)* and *question denotation* to unambiguously refer to the respective syntactic and semantic notions, and the term *question* as a cover term for both notions. This will allow us to use more natural terms in some cases, e.g., *alternative question* instead of *alternative interrogative* when talking about the syntactic object.

anti-rogative predicates based on the idea that combinations of certain semantic properties and interrogative complements result in logical triviality, which leads to unacceptability (following Gajewski 2002 and others’ idea in terms of meaning-driven unacceptability). More specifically, Theiler et al. (2019) and Mayr (2019) focus on the generalization that neg-raising predicates are anti-rogative (Zuber, 1982), and explain this generalization based on the idea that when a neg-raising predicate is composed with interrogative complements, the excluded-middle property (argued by Bartsch 1973 to derive the neg-raising inference) results in semantic triviality. Similarly, Uegaki and Sudo (2019) posit that non-veridical preferential predicates are anti-rogative, and provide an account of the pattern based on the triviality that arises from the combination of non-veridicality and preferentiality. Definitions of these semantic properties will be discussed in Section 2.1.

Although these accounts provide interesting analytical possibilities and serve as important stepping stones to addressing the overarching question on combinatorial restrictions, issues still remain with respect to the empirical status of the (hypothesized) correlations between combinatorial restrictions and lexical semantics. Indeed, White (2021) presents attested counterexamples to both of the generalizations noted above, i.e., that neg-raising predicates are anti-rogative and that non-veridical preferential predicates are anti-rogative. Furthermore, data considered in the current literature mostly come from English while the cross-linguistic status of the relevant generalizations have not been systematically investigated. This is a crucial gap in the current empirical landscape, as the theoretical accounts of the correlations between combinatorial and semantic properties cited above predict that the same correlations should hold cross-linguistically because they rely on general compositional-semantic mechanisms which are not assumed to be language-specific.

The primary goal of this paper is to improve our understanding of the cross-linguistic empirical landscape concerning the correlations between attitude predicates’ semantic and combinatorial properties. Our focus will be on non-veridical preferential predicates (NVPs), such as *hope*, *fear*, and *worry* in English as well as some of their cross-linguistic counterparts. Our point of departure is Uegaki and Sudo’s (2019) analysis of NVPs that derives the generalization that they are anti-rogative. We outline their assumptions in Section 2.1. (While it would be worth exploring analyses alternative to theirs, especially ones applying to *hope* *whether* and related cases, we leave the task for another paper.) In the following sections, we will provide counterexamples (some original and some from the literature) to Uegaki and Sudo’s (2019) generalization from English, as well as from Japanese, Mandarin, Spanish, and Turkish. These counterexamples show that Uegaki and Sudo’s (2019) account is not tenable in its original form, but they also reveal fine-grained variations in the lexical semantics of NVPs, which, to our knowledge, have not been documented in the previous literature. To preview, we will show that NVPs can be divided into three classes, varying across two factors: (i) *C(lausal)-distributivity* (i.e., the validity of the following biconditional for any question Q : $\lceil x \text{ Vs } Q \rceil$ iff there is an answer p to Q such that $\lceil x \text{ Vs that } p \rceil$; Spector and Egré 2015, Theiler et al. 2018) and (ii) *valence*, i.e., whether the predicate is evaluatively positive or negative. Among the three classes, only one class, i.e., C-distributive and positive NVPs, seems to exhibit anti-rogativity (with an important caveat about a last resort repair mechanism that involves *non-canonical composition* to be discussed below), while the other two classes—[non-C-distributive] and [C-distributive, negative]—are both compatible with interrogative complements. The empirical picture is summarized in Table 1.

	non-C-distributive	C-distributive
positive	$\checkmark V Q$, §3.1	$*V Q$ unless non-canonical composition, §4
negative		$\checkmark V Q$, §3.2

Table 1: Summary of the selectional behaviors of NVPs depending on C-distributivity and valence, i.e., evaluative positivity/negativity, with section numbers for relevant discussion.

We will argue that examining the assumptions in Uegaki and Sudo’s (2019) analysis allows us to provide theoretical explanations for these empirical observations. Specifically, we will show that two assumptions are crucial in the derivation of triviality for $NVP+Q$ in Uegaki and Sudo’s (2019) analysis: (i) NVPs are C-distributive and (ii) NVPs have a Threshold Significance Presupposition, i.e., the presupposition that there is a proposition in the set of relevant alternatives that exceeds the threshold of the attitude. We show that neither assumption holds in general. We first provide examples of non-C-distributive NVPs. Since such predicates do not meet Uegaki and Sudo’s assumption (i), their compatibility with interrogative complements

is straightforwardly expected. Furthermore, we show that evaluatively negative predicates systematically lack the Threshold Significance Presupposition, explaining the compatibility of [C-distributive, negative] NVPs with interrogative complements.²

Finally, we will note that the [C-distributive, positive] class can sometimes (appear to) take polar questions, but with an interpretation that only targets the positive answer of the question. We will suggest that these cases result from non-canonical composition between the predicate and the polar question, where the polar question is not the semantic argument of the predicate but is instead interpreted as a speech/thought report and integrated with the predicate as an adjunct.

Along the way, we will provide data from Japanese, Mandarin Chinese, Spanish, and Turkish that are consistent with the classification in Table 1. The main goal is to provide further support for existential claims about the three classes and their combinatorial patterns. That is, we provide concrete attested examples of classes that are predicted to be possible (i.e., $V + Q$ for [non-C-distributive] and [C-distributive, negative] NVPs). In addition, we provide a predictive account that maintains Uegaki and Sudo’s (2019) restriction against canonical composition between [C-distributive, positive] NVPs and interrogative clauses, and delineates the conditions under which interrogative complements will be possible with a given predicate. We also emphasize that our cross-linguistic investigation, despite being relatively small-scale, improves our understanding of the combinatorial patterns and restrictions of NVPs in several ways that would be hard to achieve if we only consider data from English (or any single language). First, there may be accidental and idiosyncratic factors that make predicates of a certain class hard to find or even non-existent in a single language. For instance, while evaluatively positive non-C-distributive predicates are hard to find in English (or Spanish and Turkish), attested examples are more readily available in Mandarin and Japanese. Second, certain constructions attested in one language may receive borderline and/or varied grammaticality judgments. If similar judgments hold for their cross-linguistic counterparts, then their borderline and variable acceptability may have a systematic explanation instead of being purely due to language-specific idiosyncratic factors. Finally, structures that cannot be distinguished at the surface level in a language may be realized differently in other languages (see, e.g., Ahn and Sauerland, 2017). Therefore, a cross-linguistic investigation can help reveal and motivate structural distinctions that are not directly observable in a particular language. For instance, the non-canonical composition analysis we propose for English (and Mandarin) *hope whether* sentences is independently motivated by similar constructions in Turkish and Japanese.

The rest of the paper is structured as follows. In Section 2, we start off the discussion by surveying Uegaki and Sudo’s (2019) analysis of the (putative) incompatibility between NVPs and interrogative clauses, identifying two crucial assumptions they made about the semantic properties of NVPs: (i) C-distributivity and (ii) Threshold Significance Presupposition. We then discuss empirical challenges for the analysis posed by White’s (2021) counterexamples and Uegaki’s (2022) experimental data. In Section 3 we propose our new classification of NVPs based on two empirical properties: (i) C-distributivity and (ii) evaluative polarity/valence (which correlates with the Threshold Significance Presupposition). In Section 3.1, we provide English as well as cross-linguistic examples of non-C-distributive predicates that are compatible with interrogative complements. In Section 3.2, we turn to C-distributive negative NVPs and note that they systematically lack the Threshold Significance Presupposition. This explains their compatibility with interrogative complements, as the presupposition is a crucial assumption in Uegaki and Sudo’s (2019) derivation of triviality for $NVP + Q$. These two classes of question-taking NVPs suggest that Uegaki & Sudo’s generalization and analysis only apply to a narrower class, i.e., C-distributive and evaluatively positive predicates such as *hope*. In Section 4, we discuss attested examples where *hope* appears to take polar interrogative clauses. After identifying three empirical properties of such examples that need to be accounted for, we suggest that such examples do not involve canonical composition between *hope* and the polar interrogative clause. We present two candidate analyses and prefer the one where such examples involve an adjunction-like mode of composition, which has been independently motivated based on related Turkish and Japanese constructions. Section 5 concludes.

²We presented this analysis at [venue redacted for anonymity] in 2022, and it was also independently developed by Klochowicz (2022).

2 Baseline generalization about non-veridical preferentials

2.1 Uegaki & Sudo (2019)

Based on examples like (2), Uegaki and Sudo (2019) (U&S) conjecture that non-veridical preferential attitude predicates are anti-rogative, i.e., that they cannot take interrogative clauses, while being able to take declarative ones. This class of predicates includes *hope*, *wish*, *expect*, and *fear*, as shown, but also *want*, *be eager*, *aspire*, *desire*, or *prefer*.

- (2) a. *Ben **hopes/wishes** which students will be invited to the party.
 b. *Chris **expects/fears** how many students will be invited to the party.
 (Uegaki and Sudo, 2019, 327, (8))

In contrast to their non-veridical counterparts, *veridical* preferential predicates—often also known as emotive factives—are responsive, i.e., they are compatible both with declarative and with interrogative clauses. Their compatibility with interrogatives is shown in (3). This class includes, but is not limited to, *be surprised*, *be annoyed*, *be glad*, *be happy*, *like*, *love*, and *hate*. We note, without attempting to say why, that some of these predicates sound better with embedded questions when these are introduced by a preposition. Prepositions that introduce questions are discussed by Mayr (2019), Egré (2008: Section 4.3), Rawlins (2013) and Uegaki and Sudo (2019: Section 5).

- (3) a. Andy **is surprised** (at/by) which students are invited to the party.
 b. Ben **is happy/glad** ?(about) which students are invited to the party.
 c. Chris **liked/hated** which students were invited to the party.
 (Uegaki and Sudo, 2019, 327, (7))

A word about what makes a predicate a (non-)veridical preferential. For a predicate *V* to be veridical, it is necessary and sufficient that sentences of the form ‘*x Vs that p*’ or ‘*x Ved that p*’ entail *p*. For a predicate *V* to be preferential, U&S hypothesize that a sufficient condition is for *V* to be focus-sensitive and gradable (Villalta 2008, Romero 2015, a.o.). Focus-sensitivity means that differences in the position of focus within *Vs that*-clause complement give rise to differences in the truth-conditions of *x Vs/Ved that p*. And gradability means that *V* can be modified by degree adverbials (*x is very surprised that p*) and participate in comparative constructions (*x is more surprised that p than y is*). Both of these properties factor into the lexical entries from U&S provided below, with the reference to a set of alternative propositions *C* to preferential predicates’ complement (focus-sensitivity), and to a comparison between degrees in the predicates’ assertion (gradability).

U&S propose to account for *why* non-veridical preferentials might be incompatible with interrogative complements. Their account rests on three general assumptions. First, embedded declaratives and interrogatives both denote objects of the same type, and attitude predicates are not restricted by their lexical entries for whether or not they are able to combine with one or the other clause type (Uegaki, 2015; Theiler et al., 2018). In the case at hand, both veridical and non-veridical preferentials will compose freely with declarative and with interrogative complements. A consequence of this assumption is that it is meaningful to compute and reason about the semantic values of (certain) sentences that are ungrammatical, like the ones in (2).

Second, semantically composing certain attitude predicates with certain clause-type denotations gives rise to meanings that are logically trivial—tautologies or contradictions—in a systematic way. We do not fully elucidate what ‘a systematic way’ means here, but U&S understand it as Gajewski’s (2002) notion of L-analyticity: A sentence has an L-analytical meaning if its meaning is tautologous or contradictory regardless of which non-logical lexical items are used to construct sentences with the same structure. In U&S’s account, given a reasonable hypothesis about their lexical semantics, the result of combining non-veridical preferentials with interrogative denotations is systematically tautologous. In contrast, composing non-veridical preferentials with declaratives and veridical preferentials with declaratives or with interrogatives avoids any logical triviality.

Finally, the grammar filters out as ungrammatical sentences that are assigned certain logically trivial meanings, in particular L-analytical ones (Barwise and Cooper, 1981; Chierchia, 2013; Gajewski, 2002). Conjoined with the second assumption, the result is that non-veridical preferentials give rise to ungrammaticality when they compose with interrogatives. In contrast, non-veridical preferentials with declaratives and

veridical ones with declaratives or interrogatives are not expected to give rise to ungrammaticality.

We elaborate now on the second one of these assumptions: How do U&S propose to define veridical and non-veridical preferential predicates, and how are tautologous meanings derived when non-veridical preferentials compose with interrogatives (and only then)?

For non-veridical preferentials, U&S assume lexical entries along the lines of (4), illustrated with *hope*_C.

- (4) $\llbracket hope_C \rrbracket^w = \lambda S_{\langle st, t \rangle} \lambda x_e: \underline{\exists p \in S[p \in C]}. \exists p' \in S \left[\begin{array}{l} p' \in C \wedge \\ \mu_w^{hope}(x, p') > \theta^{hope}(C) \end{array} \right]$
- a. $C \stackrel{\text{def}}{=}$ a contextually determined comparison class (the set of focus alternatives to p)
 - b. $\mu_w^{hope}(x, p') \stackrel{\text{def}}{=}$ the (maximum) degree to which x prefers p' at w , as it pertains to x 's hopes
 - c. $\theta^{hope}(C) \stackrel{\text{def}}{=}$ the standard threshold of preference for *hope* given a comparison class C

*Hope*_C denotes a function that takes a set S of propositions (the denotation of its declarative or interrogative clausal complement) and an attitude holder x . It is further sensitive to a comparison class C , which is the set of its clausal complement's focus alternatives. It presupposes that there is a proposition p in S that is a member of C and it asserts that there is a proposition p' in S that is a member of C^3 and (importantly) that x 's degree of preference for p' at the world of evaluation exceeds some threshold $\theta(C)$ (Romero, 2015). It is useful to note, for understanding the results below, that U&S assume that presuppositions are definedness conditions on functions (see, e.g., Heim and Kratzer 1998).

When the verb combines with a declarative, S corresponds to the singleton set $\{p\}$ where p is the proposition expressed by the declarative (Uegaki, 2015; Theiler et al., 2019). In (5), $S = \{\lambda w. \text{Kate sings at } w\}$. In this case, the comparison class C is the focus semantic value of the declarative, obtained by replacing the focused expression it contains by (relevant) expressions of the same type, that is, $C = \{\lambda w. \text{Kate sings at } w, \lambda w. \text{Lisa sings at } w, \dots\}$ (Rooth, 1992).

- (5) Ben hopes that Kate_F sings.

In the declarative case, S is a singleton and the existential quantifiers simplify. The conjunct expressing preference then requires (roughly) that the degree to which Ben prefers that Kate sings exceeds the threshold based on the degrees to which Ben prefers that anyone else sings.

When the verb combines with an interrogative as in (the ungrammatical) example (6), S is valued by the set of the interrogative's Hamblin alternatives ($S = \{p \mid \exists x[p = x \text{ sings at } w]\}$) (Hamblin, 1976; Uegaki, 2015; Theiler et al., 2019, a.o.). U&S further assume that in this case $C \subseteq S$.

- (6) *Ben hopes which student sings.

In this case, the conjunct expressing preference in (4) only requires that there be a (single) student y such that the degree to which Ben prefers that y sings exceeds the degree to which they prefer that anyone else sings. This conjunct, taken in isolation, is no problem: It predicts that the sentence should entail that there is a (single) student who Ben prefers sings.

Recall that U&S aim to derive logically trivial truth conditions for sentences like (6). To arrive at this result they appeal to a constraint associated with degree constructions which they assume to hold in general, namely the **Threshold Significance Presupposition** (TSP) stated in (7).

(7) **Threshold Significance Presupposition**

Degree constructions in general presuppose that there is an element in the comparison class whose degree along the relevant scale is higher than the threshold returned by θ .

³Degree constructions in general presuppose that the comparison term—e.g., *Johnny*, in (i)—is a member of the comparison class C —intuitively, the set of 10 year old children, in (i). See Heim (1999) on (ia); cf. Kennedy (2007), Schwarz (2010), and Solt (2011) on (ib)).

- (i) a. Among the 10 year olds, Johnny is tallest. # if Johnny is not a 10 year old.
b. For a 10 year old, Johnny is tall. # if Johnny is not a 10 year old.

The statements $\exists p \in S[p \in C]$ in (4) and (10) formulate this same presupposition for *hope* and *be happy*, modulo the higher type of λS : Some p in S must belong to the comparison class C .

Empirically, this presupposition is motivated with the example in (8). Here, the sentences in (8b) are infelicitous in the context given in (8a), which is intended to attribute indifference to John—that is, to make it such that it is not the case that John prefers any answer to the *which* question. Infelicity is expected if the sentence presupposes that there is an answer that John prefers, which is what motivates (7). If, instead, the sentence did not presuppose Threshold Significance, it should be able to mean that John has no particular dispreference for any of the propositions in the question denotation, and be felicitous (and true) in the context given.⁴

- (8) a. Context: There is no student in particular John wants to sing in an upcoming school show. He is indifferent. John has learned that Mary will sing at the show.
 b. #John {isn't happy about, doesn't like} which student will sing.
 (Uegaki and Sudo, 2019, 338, (41))

It is important to note that this presupposition is empirically motivated with veridical preferentials (*be happy, like*) for the reason that these are grammatical with embedded questions. The authors note (Uegaki and Sudo, 2019, fn. 14) that it is difficult to detect the presupposition with non-veridical preferentials like *hope*, which are nonetheless assumed to trigger it.

With preferential predicates, the ‘relevant scale’ consists of degrees of preference or dispreference (for positive and negative preferentials respectively), and the TSP can be formulated as (9).

- (9) For any $S, C \in D_{st,t}$, $x \in D_e$ and preferential verb V , $\llbracket V \rrbracket^w(S)(x)$ presupposes:
 $\exists p'' \in C[\mu_w^V(x, p'') > \theta(C)]$

Turning back to *hope*, when the predicate combines with a declarative in (5), the TSP requires that there be someone who Ben prefers sings and the assertion, that that person be Kate. These truth-conditions are neither tautologous nor contradictory. At the worlds at which it is defined, the sentence is true if Ben prefers that Kate sing, and false if they prefer someone else.

In the interrogative case in (6), the TSP requires, again, that there be someone who Ben prefers sings, but so does the assertion. When this is the case, the sentence can either be a presupposition failure, or it can be true. It can never be false. This is because all the worlds at which the sentence is defined, the sentence is true.

Given the assumption that certain sentences with systematically logically trivial truth-conditions are ungrammatical, and granted that there is no way of avoiding a tautology in the *hope wh-* case, sentences where *hope* combines with interrogative complements are then predicted to be ungrammatical. Thus U&S’s account provides a semantic explanation of the selectional restriction of non-veridical preferential predicates.

As a sanity check, let us verify that the truth conditions associated with veridical preferentials are not logically trivial regardless of whether these combine with declaratives or with interrogatives—hence, that they are not predicted to be ungrammatical. Consider the lexical entry in (10), which illustrates their assumptions for veridical preferential *be happy*. This lexical entry differs minimally from the one for *hope_C* in (4) in that it incorporates the presupposition and assertion that some p in S is true, and that the attitude holder x believes p ($\mathbf{B}_w(x, p)$).⁵

- (10) $\llbracket be\ happy_C \rrbracket^w$
 $= \lambda S_{\langle st,t \rangle} \lambda x_e : \underline{\exists p \in S[p(w) \wedge \mathbf{B}_w(x, p) \wedge p \in C]}. \exists p' \in S \left[\begin{array}{l} p'(w) \wedge \mathbf{B}_w(x, p') \wedge p' \in C \\ \mu_w^{be\ happy}(x, p') > \theta^{be\ happy}(C) \end{array} \right]$

⁴A reviewer notes that positing a Threshold Significance Presupposition is not the only way of accounting for the data point in (7), and by extension, for the anti-rogativity of NVPs. One could assume, they note, that ‘be happy’ is neg-raising(-like) in its at-issue content, with ‘ x isn’t happy that p ’ implying that ‘ x prefers not p to p .’ We agree that there are interesting alternatives to assuming the Threshold Significance Presupposition, but our more modest goal here is to present U&S’s account and the discussion of alternatives is beyond the scope of this paper.

⁵Assumptions differ as to whether emotive factives (U&S’s ‘veridical preferentials’) are veridical or factive *in addition* to presupposing that the attitude holder believes (a proposition in) their complement (Klein, 1975; Egré, 2008; Anand and Hacquard, 2014). Whether these predicates are veridical/factive or not does not affect the result that their truth-conditions avoid logical triviality. The belief presupposition, which everybody assumes, suffices for this result.

When *be happy* composes with a declarative, like in (11), we derive the presupposition that Kate sang and that Ben believes it (again because *S* is the singleton containing that proposition) and the TSP states that there is a proposition in *C* that Ben prefers over the threshold. The assertion here is strictly stronger, stating that Kate sang, that Ben believes it, *and that Ben’s preference for Kate having sung exceeds the threshold given the other propositions in C*. When defined, the sentence is true when this condition is met, and false otherwise.

(11) Ben is happy that Kate_F sang.

With an interrogative, in (12), *be happy*’s presupposition is weaker, namely that someone sang and that Ben believes that that person sang, with the TSP stating that there is a proposition in *C* that Ben prefers over the threshold. The assertion now states that someone sang, that Ben believes that that person sang, *and that their preference for that person having sung exceeds the threshold given the other propositions in C*. Again, there is no triviality here: When the sentence is defined, Ben could prefer the proposition that is true and that they believe (which makes the sentence true), or they could prefer some other proposition (which makes the sentence false).

(12) Ben is happy (about) which student sang.

In the case of *veridical* preferentials, then, the TSP is not expected to lead to trivial truth conditions with declaratives or with interrogatives, and no ungrammaticality is predicted.

Before moving on, we would like to foreshadow the fact that the lexical entries in (4) and (10) satisfy the property of C(lausal)-distributivity. We will define this property formally in Section 3.1, but at the level of intuition, this corresponds to the equivalence between *Ben is happy (about) which student sang* and *Ben is happy that *x* sang*, for some student *x*.

2.2 Empirical challenges

There have been empirical challenges to U&S’s generalization that non-veridical preferentials are anti-rogative. We review two papers where these come from. First, White (2021) provides several naturally occurring counterexamples where *hope* and *fear* take *whether*-complements, given in (13) and (14):

- (13) a. This Trump/Carson boom really has people like Bush, Walker, Rubio, and others wondering and **hoping whether** history will repeat itself and whether Republicans will return back to focusing on the establishment choices but it’s all about outsider candidates right now.
- b. I was **hoping whether** you are able to guide me [...]
- c. I have done a quite a bit of research on using a Limited Co but was **hoping whether** someone with more experience could confirm my understanding of a few points [...]
- (14) a. Interstellar space is so vast that there is no need to **fear whether** stars in the Andromeda galaxy will accidentally slam into the Sun.
- b. I **fear whether** this test would run safely on the oxygen sensor as it has a lot of drawback when compared with the others.
- c. [...] I **fear whether** I’ll have use of my arms/hands by age 55 or 60.
- d. I know parents who seriously **fear whether** their children will ever hold a meaningful job.

Prima facie, these data directly challenge U&S’s generalization. At the same time, beyond being counterexamples to the generalization, they may reveal new systematic patterns in the complementation behavior of these predicates. In Sect. 3, we will examine the properties of these counterexamples and discuss new ways to approach the relationship between the semantics of preferential predicates and their selectional properties.

In addition to White’s data, Uegaki (2022) also presents experimental results that partially challenge U&S’s generalization. In the experiment that this paper reports on, participants were presented with a set of 87 preferential predicates and rated their veridicality and compatibility with interrogative complements using a 7-point Likert scale. The experiment was done in two blocks: the veridicality block and the compatibility-with-interrogatives block. See Uegaki (2022) for details of the methodology including the procedure according to which the predicates included were selected.

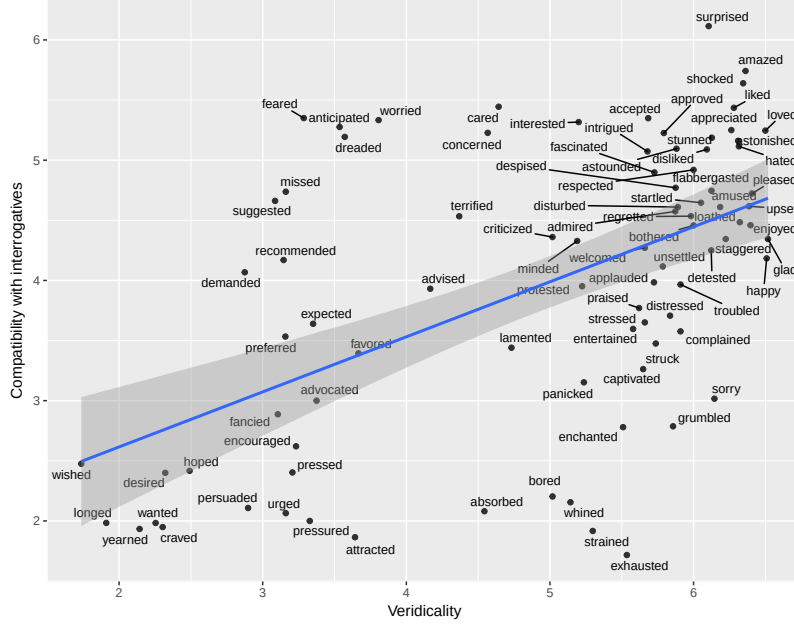


Figure 1: The plot of 87 preferential predicates in the experimental results by Uegaki (2022). The x-axis is the mean veridicality rating while the y-axis is the mean acceptability rating with interrogative complements. Mixed-effects logistic regression shows significant effect of Veridicality on Compatibility with interrogatives ($p = 0.00154$).

The results of the experiment are plotted in Fig. 1. Even though there is a correlation between veridicality and compatibility with interrogative clauses, predicates in the upper left corner constitute potential counterexamples to the generalization, as these are predicates that are judged to be compatible with interrogative complements while having low veridicality scores.

Again, in addition to being counterexamples, the observations may reveal previously unexplored systematicity. Inspecting the upper left corner of the plot, we can already make an informal observation that many of the evaluatively negative predicates, such as *fear*, *dread* and *worry* appear to have the problematic behavior. In Sect. 3, we further investigate the role of evaluative valence in the complementation pattern of preferential predicates.

2.3 Taking stock

Uegaki & Sudo (2019) propose an empirical generalization that non-veridical preferential predicates are anti-interrogative and provide an explanation for the generalization in terms of semantic triviality that results from combining non-veridical preferentials and interrogative clauses. Although a prediction of the generalization—that there is a correlation between veridicality and compatibility with interrogative clauses among preferential predicates—is experimentally confirmed by Uegaki (2022), the generalization also faces empirical challenges. As White (2021) and Uegaki (2022) show there are examples of non-veridical preferential predicates taking interrogative clauses.

Thus, the generalization as proposed by Uegaki and Sudo (2019) does not seem to stand, at least in its original form. At this point, we are left with two choices: (i) to analyze properties of counterexamples and attempt to refine the original analysis by Uegaki and Sudo (2019) and (ii) to abandon U&S’s project altogether. Although White (2021) proposes to pick the latter option, we believe that it is worthwhile to explore the first one. As we emphasized already, although the counterexamples suggest that the original generalization is invalid, they also offer intriguing data points that may shed a new light on the nature of the relationship between the semantics of preferential predicates and their selection. In the following sections, we will pursue a theory-driven investigation and scrutinize the assumptions by U&S about the semantic

properties of non-veridical preferential predicates. This leads us to identify two classes of counterexamples. Overall, we will show that a more fine-grained classification of non-veridical preferential predicates than what has been assumed in this section is possible, and that a new generalization can be stated in term of this new classification.

The resulting generalization, of course, will be open for further empirical assessment and may turn out to be incorrect. However, this does not mean that the project as a whole is not worth pursuing. Through exploring potential semantics-selection correlations and examining the counterexamples, we will advance our understanding of the lexical semantics and selectional restrictions of clause-embedding predicates.

3 An updated classification of non-veridical preferentials

Uegaki & Sudo rely on two assumptions about NVPs, namely, that they are C-distributive and that they trigger the Threshold Significance Presupposition. In this section, we show that neither assumption holds for NVPs in general, and that some of the counter-examples to U&S’s generalization can be categorized into two classes, based on whether the first or the second assumption does not hold for them. This leads to an updated classification of NVPs, which, to our knowledge, is novel. We also harness data from Mandarin Chinese, Japanese, Spanish and Turkish to provide cross-linguistic support for this classification. These data were either attested online or constructed, and judged by at least two native speakers of the corresponding language.

3.1 Non-C-distributive NVPs

Uegaki and Sudo (2019) assume, in their proposed lexical entries for NVPs, that these are C-distributive (or they would be if they were able to take interrogative clauses). A responsive predicate V is said to be *C(lausal)-distributive* iff the meanings of its interrogative- and declarative-embedding uses are related in the following way (15) (Spector and Egré, 2015; Theiler et al., 2018).

- (15) A predicate V is **C-distributive** iff for any embedded question Q , the following holds:
 $\lceil x \ V \ Q \rceil$ is true iff there is an answer p to Q (or formally, $\llbracket p \rrbracket \in \llbracket Q \rrbracket$) such that $\lceil x \ V \ p \rceil$ is true.

Most responsive predicates are C-distributive. For instance, *Al knows which player won the race* is true iff there is an answer to the embedded question, e.g., *x won the race*, such that *Al knows that x won the race*. In U&S’s derivation of their generalization, it is crucially because of this assumption that, when an NVP takes an interrogative clause, its asserted content is identical to the Threshold Significance Presupposition (whence, a logically trivial meaning and ungrammaticality).

But, not all of them are. For instance, consider Elliott et al.’s (2017) argument that *predicates of relevance* like *care* are non-C-distributive. When *care* takes a declarative clause p , it entails and presupposes that the attitude holder believes p (at least under one of its uses, see below), as shown in (16).

- (16) a. Mary cares that John left.
 $\Rightarrow$ Mary believes that John left.
b. Mary does not care that John left.
 $\Rightarrow$ Mary believes that John left.

In contrast, when *care* takes an interrogative clause Q , as in (17a), the entailment that the attitude holder believes an answer p' to Q disappears. This can be seen with the fact that sentences like (17b) are not contradictory. There, the second conjunct *wonder Q* entails that there is no answer to Q that the attitude holder believes. If the first conjunct *care Q* did entail that there was an answer p' to Q such that Mary cares that p' , that would also entail that Mary believes p' , in contradiction with the second conjunct. Therefore, C-distributivity (15) does not hold for *care*. (Of course, (17a) is *compatible* with Mary believing an answer to the question, but note that if the predicate had been C-distributive, then this would have been an entailment.)

- (17) a. Mary cares which student left.
b. Mary cares which student left and wonders which student left.

For completeness, Elliott et al.’s (2017) argument concerns the veridical/factive uses of *care*, and of predicates of relevance more generally, where this label comes from the observation that these predicates often presuppose the truth of their complement in addition to coming with the belief implication just seen. There is another use of *care* that neither implies truth nor belief (a non-emotive factive use of the predicate), which we discuss in the Appendix. This use can also be shown to be non-C-distributive, and the point that some responsive predicates are non-C-distributive stands regardless.

Given that not all predicates are C-distributive, there may in principle be NVPs that are non-C-distributive as well. When such predicates take interrogative clauses, their asserted content would no longer be predicted to be equivalent to the Threshold Significance Presupposition. Then, we would not expect the resulting meanings to be trivial and the predicates, to be ungrammatical with interrogatives. Therefore, we expect that there could be non-C-distributive NVPs that are compatible with interrogative clauses, and some attested examples in English and other languages are discussed below. For reasons that will become clear in Section 3.2, we further already divide such predicates by valence (i.e., whether they are evaluatively positive or negative).

3.1.1 Positive non-C-distributive NVPs

We first show that Mandarin Chinese *qidai* and Japanese *tanosimi-ni suru*, which can be roughly translated as *look forward to* in English, are evaluatively positive NVPs that are responsive and non-C-distributive.

Consider first the naturally occurring Mandarin sentences with *qidai* in (18) and (19), produced by a single speaker about an actor and his upcoming show. (Here and throughout, we indicate that an example comes from an online source with the superscript γ (Horn, 2018). A link to the example is then provided in the Appendix.) When *qidai* takes an interrogative clause (18), there need not be an answer p to the embedded clause such that the attitude holder wants p to be true. Rather, the attitude holder simply wants to know what the true answer is.

- (18) γ Hen qidai [ta hui zenyang quanshi zhege xinde juese].
 very QIDAI he will how interpret this new character
 “I (very much) look forward to (seeing) how he will interpret this new character.” (Mandarin)

When *qidai* takes a declarative clause (19), it entails that the attitude holder wants the declarative clause to be true, but the declarative clause itself may or may not in fact be true, i.e., *qidai* is evaluatively positive but non-veridical.

- (19) γ Ye qidai [ta neng tongguo zhege zuopin rang gengduode ren shuzhi ta].
 also QIDAI he can through this.CLF work.of.art let more people know.well him
 “I also look forward to him being able to let more people know him well through this piece of work”
 $\nRightarrow$ The actor will be able to let more people know him well.
 $\Rightarrow$ The speaker (I) wants the the actor to be able to let more people know him well. (Mandarin)

This means that *qidai*(Q) in (18) can be true without there being any answer p to Q such that *qidai*(p) is true (since *qidai*(p) entails that the attitude holder wants p). Therefore, *qidai* is non-C-distributive.

The same pattern holds for Japanese *tanosimi-ni suru*: (20a) and (21a) are naturally occurring examples of the predicate taking questions and the resulting meanings are non-C-distributive, as they do not entail the corresponding (constructed) sentences in (20b) and (21b), regardless of the choice X of the person who wins Most Valuable Player (MVP) this year or the choice Y of of the way in which Pattinson acts in The Batman.

- (20) a. γ [Kotosi-wa dare-ga MVP-o toru-no-ka] tanosimi-desu-wa.
 this.year-TOP who-NOM MVP-ACC win-NMZ-Q fun-COP.POL-PARTICLE
 “I look forward to who will win the MVP this year.”
 b. [Kotosi-wa X-ga MVP-o toru-no-ga] tanosimi-desu-wa.
 this.year-TOP X-NOM MVP-ACC win-NMZ-NOM fun-COP.POL-PARTICLE
 “I look forward to X winning the MVP this year.” (Japanese)

- (21) a. γ Watasi-wa [Pattinson-ga Batman-de nani-o suru-no-ka] tanosimi-desu.
 I-TOP Pattinson-NOM Batman-LOC what-ACC do-NMZ-Q fun-COP.POL
 “I look forward to what Pattinson will do in The Batman.”
 b. Watasi-wa [Pattinson-ga Batman-de subarasii engi-o suru-no-ga] tanosimi-desu.
 I-TOP Pattinson-NOM Batman-LOC wonderful acting-ACC do-NMZ-NOM fun-COP.POL
 “I look forward to Pattinson acting wonderfully in The Batman.” (Japanese)

A natural question arises here about whether English *look forward to* also counts as an example of an evaluatively positive responsive NVP that is non-C-distributive. Here, we remain agnostic for now: We focus on clause-embedding predicates that take finite clauses and English *look forward to* does not take *that*-clauses, and it appears that not all speakers accept the predicate with non-finite interrogative clauses either. Everybody seems to accept, for example, *look forward to seeing that/whether something is the case*. An analysis of such constructions would lead us astray of our main point, which is simply to say that positive non-C-distributive NVPs exist.

3.1.2 Negative non-C-distributive NVPs

Next, we suggest that there are also evaluatively *negative* NVPs that are responsive and non-C-distributive. Consider English *worry* with an experiencer subject. When it takes a declarative complement, the sentence is non-veridical (22a). In addition, the predicate requires that the attitude holder must consider the embedded clause an undesirable possibility (22b). That is, the predicate is evaluatively negative.

- (22) Mary worries that John didn’t go to school yesterday.
 a. $\nRightarrow$ John didn’t go to school yesterday.
 b. $\Rightarrow$ Mary considers John not going to school yesterday an undesirable possibility.

We can use the following context (23) to show that *worry* is non-C-distributive.

- (23) Context: Al’s father recently bought a new boat and is planning to visit Amsterdam with it. He is sure that there must be a place in Amsterdam where he can dock his boat, and he would be happy to dock his boat anywhere. However, for any specific place, he does not know whether it still has spots available, and he is anxiously trying to find out where exactly he can dock his boat.

In this context, (24) is true.

- (24) Al’s father worries where he can dock his boat.

However, there is no particular place Y (for any choice of Y , e.g., Dock D) such that (25) is true. This is because (25) entails that Al’s father considers being able to dock the boat at Y an undesirable possibility, which is incompatible with the context in (23).

- (25) Al’s father worries that he can dock his boat at Y .
 $\Rightarrow$ Al’s father considers him being able to dock his boat at Y an undesirable possibility.

This means that x *worries* Q can be true without there being any answer p to Q such that x *worries that* p is true, and that the predicate is (or has a use that is) non-C-distributive.

About the predicate English *worry*, we will further note the following. First, when the predicate occurs in different syntactic frames (e.g., with a clausal or expletive subject, like in *That John didn’t go to school yesterday worries Mary* or *It worries Mary that John didn’t go to school*), or when the related deverbal adjectival construction with *be worried* is used, a veridical reading becomes available. We set this aside but refer the reader to Hartman (2012) for further discussion.

Second, when *worry* takes an interrogative clause, there might be a general preference for using the preposition *about* to introduce it. This might worry some readers about the possibility that, in the examples above, the predicate might not be composing directly with a clausal argument: The preposition might be playing a non-trivial role either in the syntax or the semantics. But, the naturally occurring examples in (26), where *worry* occurs preposition-less with interrogative clauses should serve to alleviate the worry. At

least some English speakers do not require prepositions, and as will suggest shortly, these are not required cross-linguistically.

- (26) a. γ Mr. Nelson denies he is “a real collector – the kind of person who buys first and then worries where he can find room for his purchases.”
 b. γ As Ashley hands his church keys back he worries where he can spend the night.
 c. γ I have a sister in California that worries where she can live because she is white and her kids are mixed race.
 d. γ Lucas tells Sami that he loves her and Sami tells him that she loves him too. Sami worries what she can do since she loves them both.

Cross-linguistically, Mandarin *danxin* (27), Japanese *sinpai-suru* (28), Spanish *preocupar* (29), Turkish *endişelen-* (30), which are close translations of English *worry*, all similarly exhibit non-C-distributivity: the (a)-sentences can be true without the (b)-sentences being true for any Y .^{6,7}

- (27) a. Wode baba hen danxin [ta neng zai nali ting tade chuan].
 My father very worry he can at where dock his boat
 “My father worries (very much) about where he can dock his boat.”
 b. Wode baba hen danxin [ta neng zai Y (nali) ting tade chuan].
 My father very worry he can at Y there dock his boat
 “My father worries (very much) that he can dock his boat at Y .” (Mandarin)
- (28) a. Kare-wa [doko-ni hune-o tome-rareru-ka-o] sinpai-siteiru
 he-TOP where-LOC boat-ACC park-can-Q-ACC worry-do.ASP
 “He worries where he can dock his boat.”
 b. Kare-wa [Y -ni hune-o tome-rareru-no-o] sinpai-siteiru
 he-TOP Y -LOC boat-ACC park-can-NMZ-ACC worry-do.ASP
 “He worries that he can dock his boat at Y .” (Japanese)
- (29) a. A mi padre le preocupa [dónde puede / va a atracar Antonio el barco].
 To my father CL worries where could.IND / goes.IND to dock Antonio the boat
 “My father worries about where Antonio can / is going to dock his boat.”
 b. A mi padre le preocupa [que Antonio pueda / vaya a atracar el barco en Y].
 To my father CL worries that Antonio can.SUBJ / goes.SUBJ to dock the boat at Y
 “My father worries that Antonio can / is going to dock the boat at Y .” (Spanish)
- (30) a. Babam [gemisini nereye park ed-eceğ-in-den] endişeleniyor.
 my dad his boat where park do-NMZ.FUT-3S.POSS-ABL worry
 “My dad is worrying/worried where he’ll park his boat.”
 b. Babam [gemisini Y -ye park ed-eceğ-in-den] endişeleniyor.
 my dad his boat Y -LOC park do-NMZ.FUT-3S.POSS-ABL worry
 “My dad is worrying/worried that he’ll park his boat in Y .” (Turkish)

Other contexts, like the one in (31), can be imagined that also bring out natural non-C-distributive uses of such predicates.

- (31) Context: Sam is invited to an event. Not knowing much about it, they do not know what type of clothes would be appropriate: Casual, business casual, formal? They possess all relevant types of clothes and are happy to wear anything as long as it is appropriate for the occasion.

While (32a) can be true in this context, given that Sam is happy to wear anything, (32b) is false.

⁶Although some authors classify Spanish *preocupar* ‘worry’ as factive emotive (e.g., Villalta 2008), (29b) crucially allows for a non-veridical interpretation of the *that*-clause. The difference in embedded mood between (29a) and (29b) is due to the fact that embedded interrogative clauses obligatorily appear in the indicative while *that*-clauses under *preocupar* appear in the subjunctive per default —though see Falkner (2022) for special discourse effects triggered by the indicative.

⁷The Turkish translation (30) uses the future instead of a possibility modal in the embedded clause because (for reasons that we do not know) the latter sounds less natural. Despite this difference, our crucial point remains, i.e., the predicate is non-C-distributive because the (a) sentence can be true without the (b) sentence being true for any place Y .

- (32) a. Sam worries what they should wear.
b. Sam worries that they should wear casual/business casual/formal.
The examples below suggest the same conclusion in Mandarin, Japanese, Spanish and Turkish.
- (33) Yuehan hen danxin [ta yao chuan shenme].
John very worry he should wear what
“John worries (very much) what he should wear.” (Mandarin)
- (34) John-wa [nani-o kiru-beki-ka-o] sinpai-siteiru
John-TOP what-ACC wear-should-Q-ACC worry-ASP
“John worries what he should wear.” (Japanese)
- (35) A Juan le preocupa [qué debería ponerse].
to Juan CL worries what should.3SG wear
“It worries John what he should wear.” (Spanish)
- (36) John [ne giy-me-si gerek-tiğ-in-den] endişeleniyor.
John what wear-NMZ-3S.POSS MOD-COP.NMZ-3S.POSS-ABL worries
“John is worried what he should wear.” (Turkish)

3.2 Negative NVPs do not trigger the Threshold Significance Presupposition

Our second class of question-embedding NVPs consists of predicates such as *fear*, which are *evaluatively negative* and C-distributive. Recall that in Uegaki and Sudo’s formal derivation of the incompatibility between NVPs and embedded questions, the second crucial assumption is that NVPs have a Threshold Significance Presupposition, whose definition (7) is repeated below in (37).

- (37) **Threshold Significance Presupposition**
Degree constructions in general presuppose that there is an element in the comparison class whose degree along the relevant scale is higher than the threshold returned by θ .
(Uegaki and Sudo, 2019, 335)

According to their analysis, when a NVP takes an embedded question, the resulting meaning is trivial because the asserted content is equivalent to the Threshold Significance Presupposition. In the previous section, we saw that the equivalence does not hold for non-C-distributive predicates, which means such predicates are compatible with embedded questions because no triviality arises. Now, if there are predicates that do not have the Threshold Significance Presupposition to begin with, the equivalence will also fail to hold (even if they are C-distributive). That is, no triviality will arise when such predicates take embedded questions. In this section, we show that evaluatively negative predicates such as *fear* are such predicates.

Uegaki and Sudo’s empirical motivation for the Threshold Significance Presupposition is based on *evaluatively positive* predicates such as *like* and *be happy*, e.g., (38).

- (38) Context: It is common knowledge that John knows which student will sing. Al knows that there is no particular student John wants to sing (i.e., John is indifferent about which student will sing).
Bo asks Al: “How does John feel about which student will sing?”
a. Al: #John doesn’t like which student will sing.
b. Al: #John isn’t happy about which student will sing.
(adapted from Uegaki and Sudo 2019, 338, (41))

If there were no Threshold Significance Presupposition, Al’s answers in (38) would simply mean that it is not the case that John likes/is happy about which student will sing, and should be true in this context. However, empirically, such answers seem to presuppose the existence of a student John prefers to sing and, consequently, they feel infelicitous in the context where John does not have any preference. This point is also illustrated by (39).

- (39) Context: Same as (38).
Bo: Is John happy about which student will sing? / Does John like which student will sing?
Al: #No.

However, for evaluatively negative predicates such as *hate* and *be upset*, Al’s answers in (40) and (41) seem fine in the corresponding context where the Threshold Significance Presupposition is not met.

- (40) Context: It is common knowledge that John knows which student will sing. Al knows that there is no student such that John will hate it or be upset if they will sing.
Bo asks Al: “How does John feel about which student will sing?”
a. Al: John doesn’t hate which student will sing.
b. Al: John isn’t upset about which student will sing.
- (41) Context: Same as (40).
Bo: Is John upset about which student will sing? / Does John hate which student will sing?
Al: No.

We take the contrast between evaluatively positive and negative predicates to suggest that evaluatively negative predicates in fact do not have a Threshold Significance Presupposition (see also Klochowicz 2022). Without this presupposition, U&S’s formal analysis predicts that for an evaluatively negative NVP such as *fear*, *A fears Q* simply means *for some p in Q, A fears that p*, which is not a trivial meaning. Therefore, we no longer expect negative NVPs to be incompatible with embedded questions.

This prediction of the revised U&S’s analysis is borne out. Regarding polar interrogative complements, White (2021) provides the following corpus examples (42), repeated from (14).

- (42) a. Interstellar space is so vast that there is no need to **fear whether** stars in the Andromeda galaxy will accidentally slam into the Sun.
b. I **fear whether** this test would run safely on the oxygen sensor as it has a lot of drawback when compared with the others.
c. [...]I **fear whether** I’ll have use of my arms/hands by age 55 or 60.
d. I know parents who seriously **fear whether** their children will ever hold a meaningful job.

We can see that *fear whether p* is compatible with a scenario where the attitude holder fears that *p*, e.g., (42a), as well as scenarios where the attitude holder fears that $\neg p$, e.g., (42b)–(42d).

And below are some naturally-occurring examples of *fear* taking constituent questions (43).

- (43) a. $\neg$ Both women initially worried about taking the bus, **fearing who else** might be riding the bus. Both have been pleasantly surprised. Express bus seats recline, and have individual reading lights and air-conditioning vents.
b. But if my party doesn’t enter this city, I **fear who else** will.

Martha Wells, 2016, *The Edge of Worlds* chapter 2; Start Publishing

We can verify that *fear* indeed does not have a Threshold Significance Presupposition when it takes questions. For instance, (44) is compatible with a scenario where there exists no person *x* such that the women fear that *x* might be riding the bus. Contrast this case with negated *be happy* and *like* in (38).

- (44) The women do not fear who else might be riding the bus.

Cross-linguistically, there are further examples of evaluatively negative NVPs that can embed questions. For example, consider Turkish *kork-* “fear” (45).

- (45) John [komsusunun haftasonu evde ol-up ol-ma-yacağ-in-dan] korkuyor.
John his neighbor weekend at home be-CONJ be-NEG-NMZ.FUT-3S.POSS-ABL fear
“John fears whether his neighbor will be home over the weekend.” (Turkish)

This sentence is felicitous in the GOODFRIEND context (46), where John fears that his neighbor will not be home, as well as the NOISEHATER context (47), where John fears that his neighbor will be home.

- (46) GOODFRIEND context: John and his neighbor are good friends. His neighbor is away on a business trip, which may be long or short. John wants to throw a party this weekend.
- (47) NOISEHATER context: John likes to invite his friends over to party, but his neighbor hates noise and they have had several heated quarrels. His neighbor is away on a business trip, which may be long

or short. John wants to throw a party this weekend.

The predicate is also compatible with constituent questions (48).

- (48) John [maçı hangi oyuncunun kazan-acağın-dan] korkuyor
 John match which player win-NMZ.FUT-3S.POSS-ABL fear
 “John fears which player will win the match.” (Turkish)

We can verify that it does not have a Threshold Significance Presupposition: (49) does not presuppose that there exists a player x such that John fears that x will win the match.

- (49) John [maçı hangi oyuncunun kazan-acağın-dan] korkmuyor
 John match which player win-NMZ.FUT-3S.POSS-ABL fear.NEG
 “John does not fear which player will win the match.” (Turkish)

The same patterns hold for Japanese *osore(ru)* “fear.”

- (50) John-wa [rinzin-ga konsyuumatu ie-ni iru ka-o] osore-teiru
 John-TOP neighbor-NOM this.weekend home-LOC be Q-ACC fear-ASP
 “John fears whether his neighbor will be home.” (Japanese)

The sentence is felicitous in the GOODFRIEND context as well as the NOISEHATER context. And the predicate is also compatible with constituent questions (51).

- (51) John-wa [dono sensyu-ga sono siai-de katu-ka] osore-teiru
 John-TOP which player-NOM the match-LOC win-Q fear-ASP
 “John fears which player will win the match.” (Japanese)

We can verify that it does not have a Threshold Significance Presupposition: (52) does not presuppose that there exists a player x such that John fears that x will win the match.

- (52) John-wa [dono sensyu-ga sono siai-de katu-ka] osore-teinai
 John-TOP which player-NOM the match-LOC win-Q fear-ASP.NEG
 “John does not fear which player will win the match.” (Japanese)

We emphasize that the revised U&S’s analysis, which drops the Threshold Significance Presupposition for evaluatively negative NVPs, does not predict that they will always be compatible with interrogative clauses. It only says that, if they were to take interrogative clauses, the resulting meanings would not be trivial. However, there can be various other factors in a language that make such predicates incompatible with some or perhaps even all interrogative clauses.

For instance, while Spanish *temer* “fear” is compatible with constituent questions (53), it sounds quite bad with polar questions (54).

- (53) Juan teme [qué puede pasar si su vecino está en casa].
 Juan fears what can.IND happen if his neighbor is at home
 “Juan fears what can happen if his neighbor is at home” (Spanish)

- (54) ???Juan teme [si su vecino estará en casa].
 Juan fears whether his neighbor will.be at home
 “Juan fears whether his neighbor will be at home” (Spanish)

Mandarin *haipa* “fear” exhibits the opposite pattern. It is compatible with polar questions just like its English, Turkish and Japanese counterparts (55).

- (55) Yuehan (hen) haipa [tade linju huibuhui zaijia].
 John very fears his neighbor will.not.will at.home
 “John fears (very much) whether his neighbor will be at home” (Mandarin)

In contrast, *haipa* cannot take a constituent question. The only available interpretation of (56) is one where

the wh-phrase is interpreted as an indefinite. In this case, the embedded clause is a declarative clause rather than a constituent question.⁸

- (56) Yuehan (hen) haipa [naming xuanshou hui yingde bisai].
 John very fears which.CLF player will win match
 Intended, unavailable: “John fears which player will win the match.”
 Available: “John fears that some player will win the match.” (Mandarin)

Explaining why the Spanish and Mandarin counterparts of *fear* are incompatible with some types of interrogative clauses is beyond the scope of this paper.⁹ For our purposes here, the fact that these predicates are compatible with *some* interrogative clauses provide further evidence for the revised U&S’s analysis, according to which evaluatively negative predicates taking interrogative clauses will not result in semantic triviality.

3.3 Summary of the updated classification of NVPs

In this section, we identified two relevant factors influencing a NVP’s compatibility with embedded questions: (i) whether the predicate is C-distributive, and (ii) the valence of the predicate (which correlates with whether the predicate has a Threshold Significance Presupposition). This leads to an updated classification of NVPs, summarized in Table 2, which contains two classes of responsive NVPs in addition to the class of anti-rogative NVPs predicted by Uegaki and Sudo (2019).

	Lexical semantics		Combinatorial	
Class	C-distr.	Valence	Canonical composition with Q	Examples
1	no	positive	✓	Mandarin <i>qidai</i> , Japanese <i>tanosimi</i>
		negative	✓	English <i>worry/be worried</i> , Japanese <i>sinpai-suru</i> , Mandarin <i>danxin</i> , Spanish <i>preocupar</i> , Turkish <i>endişelen-</i>
2	yes	negative	✓	English <i>fear</i> , Japanese <i>osore(ru)</i> , Turkish <i>kork-</i>
3	yes/NA	positive	*	Japanese <i>nozomu</i> , Turkish <i>um-</i> , Spanish <i>esperar</i> English <i>hope</i> , Mandarin <i>xiwang</i>

Table 2: Summary of our updated classification of NVPs.

First, for NVPs that are non-C-distributive, regardless of whether they have a Threshold Significance Presupposition when they take questions, the asserted content is not equivalent to it and hence no triviality arises. Therefore, U&S’s original analysis does not apply to such predicates, and we expect to find non-C-distributive predicates that can take embedded questions. Examples of evaluatively positive question-taking NVPs that are non-C-distributive include Mandarin *qidai* and Japanese *tanosimi*, which can be roughly translated as English *look forward to*. Examples of evaluatively negative responsive NVPs that are non-C-distributive include English *worry* (with an experiencer subject) and its close translations in Japanese (*sinpai-suru*), Mandarin (*danxin*), Spanish (*preocupar*), and Turkish (*endişelen-*).

Second, for NVPs that are evaluatively negative, we showed that they do not have a Threshold Significance Presupposition. As a result, such predicates taking interrogative clauses would not lead to trivial meanings.

⁸We can further illustrate the unavailability of the embedded question interpretation by considering sentences that do not have a wh-indefinite interpretation. For instance, (i) is totally unacceptable.

- (i) *Yuehan haipa [tamen tigong na xie yinliao].
 John fears they serve which CL.pl beverage
 *wh-Q: “John fears which kinds of beverage they serve.”
 *wh-indefinite: “John fears that they serve some kinds of beverage.” (Mandarin)

⁹The incompatibility might be due to purely syntactic constraints that are language-specific, and in principle we do not rule out the possibility that it might be due to subtle meaning differences between English *fear* (and its Turkish and Japanese counterparts) on the one hand and the Spanish and Mandarin counterparts on the other. However, so far we have not detected any meaning difference between these predicates from different languages that would tease them apart.

Therefore, U&S’s original analysis does not apply to such predicates, and we expect to find a class of question-taking NVPs that are evaluatively negative and C-distributive. Examples of such predicates include English *fear* and its close translations in Japanese (*osore(ru)*) and Turkish (*kork-*).

Finally, for positive NVPs that are C-distributive, U&S’s original analysis predicts that they are incompatible with embedded questions due to the asserted content being equivalent to the Threshold Significance Presupposition.¹⁰ We maintain their analysis for this class of predicates, which includes English *hope* and its cross-linguistic close translations. Note that our classification here only concerns cases of what we call canonical composition with questions, where the predicate takes the canonical, symmetric denotation of its embedded question as its semantic argument. In the next section, we will discuss cases where a Class 3 predicate appears to take an embedded question, and argue that they involve non-canonical composition.

4 Non-canonical composition with embedded questions

4.1 Positive NVPs apparently compatible with embedded questions

This section concerns cases where positive NVPs appear to take embedded questions. We start with the English predicate *hope*, a paradigmatic example of a positive NVP. The sentences in (57), repeated from (13), are corpus data presented by White (2021) showing that *hope* seems to be able to take polar question complements, contrary to Uegaki and Sudo’s (2019) empirical generalization that *hope* cannot take questions.¹¹

- (57) a. I was **hoping whether** you are able to guide me[...]
 b. I have done quite a bit of research on using a Limited Co but was **hoping whether** someone with more experience could confirm my understanding of a few points[...]
 c. This Trump/Carson boom really has people like Bush, Walker, Rubio, and others wondering and **hoping whether** history will repeat itself and whether Republicans will return back to focusing on the establishment choices but it’s all about outsider candidates right now.

White, a native speaker of English, reports that he finds these examples perfectly acceptable in context. He acknowledges, however, that not all native speakers accept them. The native speakers of English we consulted generally found these examples at least somewhat acceptable, and found the intended interpretations clear in context: Sentence (57a) is interpreted as the speaker asking whether the addressee is able to help them while hoping that they are able to do so, which is effectively a polite way to request for help. Sentence (57b) has a similar interpretation. Finally, (57c) means that people like Bush and others are wondering whether Republicans will return back to focusing on the establishment choices and hoping that they will. However, many noted that *hoping whether* is an unusual/marked construction that they personally would not normally use.¹²

Now, given that not all native speakers of English find these sentences acceptable (and in fact, many find these somewhat marked), one natural question is whether such sentences are truly grammatical or whether

¹⁰Given that such predicates cannot take interrogative clauses, they are only presumed to be C-distributive according to U&S’s original analysis based on the fact that most responsive predicates are C-distributive. This is why the C-distributivity of this class is marked as NA/yes in Table 2.

¹¹In this paper, following previous works, we focus on *whether*-interrogatives and set aside any further differences they may have from *if*-interrogatives. But we note that *if*-interrogatives under *hope* are also attested. Below are some examples from the same two corpora from which White (2021) extracted the *hope whether* examples.

- (i) a. I was hoping if I could just ask you a few questions about the disappearance of John Dexter.
 b. I was hoping if you could tell me about the cannacig or their cumulus.

To the best of our knowledge, *hope if* and *hope whether* have very similar interpretations. In principle we can apply our analysis of *hope whether* to these *hope if* examples. But we note that Eckardt’s (2007) analysis of *if*-interrogatives, which is similar to our analysis in Section 4.3 in certain respects, may provide an alternative derivation that is specific to *hope if* (since her analysis is about how *if*-interrogatives are different from *whether*-interrogatives) but that ends up predicting an interpretation of *hope if* which is the same as that of *hope whether* under our analysis. A detailed comparison between these two possibilities is beyond the scope of this paper and we leave it for future research.

¹²We consulted 5 native English speakers. 2 consultant found these examples perfectly acceptable, 1 found them unacceptable, and the other 2 found them acceptable/interpretable in context but marked to various degrees. Furthermore, every time we presented this work, there were some native speakers in the audience expressing discomfort with these examples.

they are merely acceptable in context due to the interpreter’s accommodation of production errors. For the purpose of this paper, we remain largely agnostic about this question. Instead, we identify the following empirical properties of sentences where *hope* takes polar questions and aim to provide an analysis that accounts for them.

- (58) Empirical properties of examples of *hope* taking polar questions:
- a. **Marginal/Variable Acceptability:** Many speakers find such examples at least somewhat marked, and some even find them totally unacceptable.
 - b. **Interpretive Asymmetry:** *x hopes whether p* is only compatible with a context where *x* hopes that *p*, but not one where *x* hopes that $\neg p$.
 - c. **Inquisitive Implication:** *x hopes whether p* is not simply interpreted as *x hopes that p*. Rather, it also implies that *x* wonders whether *p*.

Let us go through these empirical properties in more detail. Regarding (58a), as already noted above, even though there are attested examples of the English predicate *hope* taking questions (57), many native speakers find such examples at least somewhat marked, and some even find them totally unacceptable. Later we will show that this also holds for the counterparts of *hope* in other languages that we surveyed.

But before that, let us first take a closer look at the interpretations of the English examples in (57). These, and other examples of *hope whether p* that we have come across all display an interesting interpretive pattern: To the extent that they are acceptable, they may all be roughly paraphrased as *hope that p* but never as *hope that not p*. Sentence (57a), for example, implies (59a) and is incompatible with (59b).

- (59) a. I was hoping that you’re able to guide me.
b. I was hoping that you aren’t able to guide me.

The *hope that not p* paraphrase remains absent even when we construct contexts rich enough to support that interpretation. Consider again the GOODFRIEND and NOISEHATER contexts from Section 3.2, repeated below. The GOODFRIEND context motivates and attributes to John a positive preference for his neighbor to be home, and the NOISEHATER context does the opposite.

- (60) GOODFRIEND context: John likes to invite his friends over to party. He and his neighbor are good friends. His neighbor is away on a business trip, which may be long or short. John wants to throw a party this weekend.
- (61) NOISEHATER context: John likes to invite his friends over to party, but his neighbor hates noise and they have had several heated quarrels. His neighbor is away on a business trip, which may be long or short. John wants to throw a party this weekend.

We find that a sentence like (62a) is only judged true in the GOODFRIEND context, and that it is judged false in the NOISEHATER context. This is the same pair of judgments that the *hope that p* paraphrase in (62b) receives, and the opposite of what the *hope that not p* paraphrase in (62c) receives.

- (62) a. John is hoping whether his neighbor will be home this weekend.
GOODFRIEND: True; NOISEHATER: False.
- b. John is hoping that his neighbor will be home this weekend.
GOODFRIEND: True; NOISEHATER: False.
- c. John is hoping that his neighbor won’t be home this weekend.
GOODFRIEND: False; NOISEHATER: True.

The pair in (63) provides additional support for the same point: While (63a) is an acceptable way of expressing one’s desire to leave, (63b) is not.

- (63) Alex and Sam are at a party together. Alex says to Sam:
- a. I’m feeling tired. I was hoping whether we could leave.
 - b. I’m feeling tired. #I was hoping whether we have to stay.
Intended: I was hoping that we didn’t have to stay/that we could leave.

These observations strongly suggest that a *hope that p* paraphrase is available for *hope whether p*, but that a *hope that not p* paraphrase is consistently unavailable. If alternative ways of interpreting *hope whether p* exist, be it as *hope that not p* or as a non-propositional relationship to the interrogative, e.g., see *worry* in Section 3.1.2, we have not found evidence for them.

Furthermore, we note that *hope whether p* only has *hope that p* as a *rough* paraphrase. Our consultants reported that *hope whether p* not only expresses a preference for *p* but also conveys that the attitude holder is actively entertaining the question of whether *p*. This Inquisitive Implication (58c) of *hope whether p* is another empirical property that any satisfactory analysis of this construction should account for.

Cross-linguistic data The counterparts of *hope* in other languages also vary in their acceptability with polar interrogatives. We illustrate with data from Mandarin, Spanish, Turkish and Japanese. Mandarin is the only language in the sample where we have found that such constructions are acceptable (by some but not all native speakers), and their interpretation patterns with the English data discussed above.¹³

In Mandarin, the lexical material inside a polar interrogative affects whether it can be embedded under *xiwang*, “hope.” This is shown by the contrast between (64a) and (64b). We do not have a general characterization of what conditions such differences in acceptability, but note that the question *whether x can help y*, adapted from ex. (57a), also leads to improvement in English. The acceptability of sentences like (64b), however, is subject to across speaker variation.¹⁴

- (64) a. *Yuehan xiwang [(daoshihou) tade linju huibuhui zai jia].
 John hope then his neighbor will.not.will at home
 Intended: “John hopes whether his neighbor will be at home then.”
 b. %Yuehan xiwang [(daoshihou) tade linju nengbuneng bangbang ta].
 John hope then his neighbor can.not.can help him
 Intended: “John hopes whether his neighbor can help him.” (Mandarin)

For speakers who find examples like (64b) acceptable, the sentence entails that John hopes that his neighbor can help him, and is incompatible with contexts in which he hopes that his neighbor can’t help him. The sentence also implies that John is wondering whether his neighbor can help him, i.e., it also has an inquisitive implication. That is, the empirical properties of English *hope whether* sentences also extend to their Mandarin counterparts.

Spanish has two strategies for finite interrogative embedding, illustrated in (65a) and (65b) with the verb *decir*, “say.” A choice is available only for some clause-embedding predicates and it gives rise to an interpretive difference. Bare *si* interrogatives under *decir* require that the matrix subject tell an answer to the question, e.g., that the neighbor could help, in (65a). In contrast, *que si* interrogatives under the same predicate require that the matrix subject raise the issue expressed by the interrogative complement, i.e., that Juan *asks* whether the neighbor can help (Suñer, 1993). (We illustrate with polar interrogatives, but the same pattern affects *wh*- questions as well.)

- (65) a. Juan les dijo [si el vecino podía ayudar].
 Juan them-DAT said whether the neighbor could.IND help
 “Juan told them whether the neighbor could help.”
 b. Juan les dijo [que si el vecino podía ayudar].
 Juan them-DAT said QUE whether the neighbor could.IND help
 “Juan asked them whether the neighbor could help.” (Spanish)

The examples in (66a) and (66b) illustrate that both options give rise to ungrammaticality with *esperar*, “hope.” Note that we are using an embedded interrogative which we know gives rise to acceptable results

¹³Some languages may allow for the same attitude verb to compose with embedded clauses that differ in their syntax, which in turn may give rise to differences in acceptability and interpretation. This point is not relevant for Mandarin *hope*, but discussed for Spanish here, and for Japanese and Turkish in section 4.3.

¹⁴The author of the paper who is a native speaker of Mandarin finds (64b) perfectly acceptable. We consulted 4 additional native speakers. One consultant similarly found it perfectly acceptable, one found it highly marked, and the other two found it somewhat marked. Furthermore, there have been about 10 native speakers of Mandarin at several presentations of earlier versions of this work. Some found (64b) perfectly acceptable, while others found it marked to various degrees.

in English and Mandarin, and that we have not found evidence for across speaker variation here. All of the speakers that we have asked have rejected sentences of the form (66).

- (66) a. *Espero [si el vecino me puede ayudar].
hope-1SG whether the neighbor me-ACC can.IND help
Intended: “I hope / was hoping whether the neighbor can help me.”
b. *Espero [que si el vecino me puede ayudar].
hope-1SG QUE whether the neighbor me-ACC can.IND help
Intended: “I hope / was hoping whether the neighbor can help me.” (Spanish)

Japanese and Turkish, which pattern similarly for the case at hand, also feature different types of embedded clauses. *Nominalized* embedded interrogatives are unacceptable under *nozomu* and *um-*, “hope,” in Japanese and Turkish. Here, we see again that the choice of lexical material that led to improvement in English and Mandarin is not enough here.

- (67) a. *Taro-wa [tomodati-ga sonotoki ie-ni iru-ka-o] nozon-deiru
Taro-TOP friend-NOM then home-LOC be-Q-ACC hope-ASP
Intended: “Taro hopes whether his friend will be at home then.”
b. *Taro-wa [tomodati-ga mondai-o tokeru-ka-o] nozon-deiru
Taro-TOP friend-NOM problem-ACC solve.can-Q-ACC hope-ASP
Intended: “Taro hopes whether his friend can solve the problem.” (Japanese)
(68) a. *[Bana yardımcı ol-up ol-ama-yacağ-ımız-ı] um-uyor-du-m.
1S.DAT helper be-COORD be-NEG.ABIL-FUT-2P-ACC hope-IPFV-PST-1S
Intended: “I was hoping whether you could help me.”
b. *[Birisi-nin yazdıklarımı teyit ed-ip et-me-yeceğ-in-i] um-uyor-du-m.
someone-GEN what I wrote confirmation do-COORD do-NEG-FUT-3S-ACC hope-IPFV-PST-1S
Intended: “I was hoping whether someone could confirm what I wrote.” (Turkish)

The second type of embedded questions that is relevant for both languages involves clauses that are not nominalized and that are introduced by the morphemes *to* (Japanese) and *diye* (Turkish). With *diye* and *to* clauses, it does become possible to compose “hope” with embedded interrogatives. Because there is reason to think that such clauses have syntactic and semantic properties that set them apart from the embedded clauses that we have presented so far in English, Spanish and Mandarin, we present the relevant data and discussion in Section 4.3.

Interim summary and outlook We have identified three empirical properties of sentences where *hope* (and its cross-linguistic counterparts) appears to take questions: Marginal/Variable Acceptability (58a), Interpretive Asymmetry (58b), and Inquisitive Implication (58c). Regardless of one’s position on the grammaticality status of *hope whether* sentences, any satisfactory analysis should account for these properties.

In light of these properties, taking an extreme position about the *hope whether* sentences is unlikely to provide a satisfactory analysis. Those who think there is nothing special about *hope* taking questions need to provide a semantics of *hope* such that the result of canonical composition with a question will lead to Interpretive Asymmetry and an Inquisitive Implication. To our knowledge, no such an account has been provided. Meanwhile, for those who think that *hope whether* sentences are simply ungrammatical and all the attested examples are just production errors, the main challenge would be to account for Interpretive Asymmetry: If listeners/readers of *hope whether p* are simply trying to find a contextually appropriate interpretation to accommodate the production error, why can they not interpret it as *hope that ¬p* even when such an interpretation is supported by the context?

In the rest of this section, we present two candidate analyses. What they have in common is that they both assume *hope whether p* sentences involve *non-canonical composition* between *hope* and the embedded question. That is, in such sentences *hope* does not take the canonical, symmetric/bipolar denotation $\{p, \neg p\}$ of a polar question as its semantic argument. Assuming that non-canonical composition can only be used as a last resort when canonical composition yields trivial interpretations and its availability is subject to variation, both analyses can account for the Marginal/Variable Acceptability of *hope whether* sentences. The

two analyses differ in terms of the concrete non-canonical composition mechanism involved. In Section 4.2, we present the first candidate analysis. It is based on the independently motivated treatment in the literature according to which *whether p* has *p* as its *highlighted content*, and it further assumes that in *hope whether p* the composition is non-canonical in that *hope* is composed with the highlighted content. We discuss issues with this analysis despite its initial appeal. We then present our preferred analysis in Section 4.3, according to which the interrogative clause in *hope whether p* is understood as a speech/thought report and then further integrated with the predicate *hope* as an adjunct.

4.2 Candidate analysis 1 (ultimately dispreferred): highlighting

In the previous section, we observe that in *hope whether p*, the attitude only targets the positive answer *p*. One natural strategy, then, is to somehow reduce *x hope whether p* to *x hope that p*. However, as White (2021) points out, it will not do to say that the complementizer *whether* is simply replaced by, or that it is sometimes the pronunciation of, the complementizer *that*. This is because examples like (57c) are attested, where *hope* can be conjoined with *wonder* and then combined with a *whether* clause. If this *whether* clause were secretly a declarative, it would be incompatible with *wonder*, which does not take declarative complements.

One candidate analysis that avoids this issue, which we ultimately disprefer despite its initial appeal, is that in sentences of the form *x hope whether p*, the polar question has *p* as its *highlighted content* (Roelofsen and van Gool, 2009; Roelofsen and Farkas, 2015; Theiler, 2021), and the predicate *hope* targets this highlighted content. Concretely, this analysis assumes that interrogatives have two meaning dimensions: their ordinary and their highlighted content. For a polar interrogative *whether p*, its *ordinary* content $\llbracket \cdot \rrbracket^o$ is the familiar symmetric denotation $\{p, \neg p\}$, whereas its *highlighted content* $\llbracket \cdot \rrbracket^h$ is the singleton set $\{p\}$ containing the denotation of the radical, in (69). We set aside the question of whether a similar result could be obtained in monopolar approaches to polar interrogative denotations, as in Biezma and Rawlins (2012).

$$(69) \quad \begin{aligned} \llbracket \text{whether } p \rrbracket &= \langle \llbracket \text{whether } p \rrbracket^o, \llbracket \text{whether } p \rrbracket^h \rangle \\ &= \langle \{p, \neg p\}, \{p\} \rangle \end{aligned}$$

Canonically, clause embedding predicates compose with the ordinary content of interrogatives. In the case of *hope whether*, however, this analysis assumes that *hope* non-canonically composes with the highlighted content of interrogatives to give rise to truth conditions along the lines of (70). Here, assume that hope_C^{+H} is defined on the template of hope_C , in (4), except that its first argument is provided by the highlighted content.

$$(70) \quad \begin{aligned} \llbracket \text{hope}_C^{+H} \text{ whether } p \rrbracket &= \llbracket \text{hope}_C^{+H} \rrbracket^w(\llbracket \text{whether } p \rrbracket^h) \\ &= \llbracket \text{hope}_C^{+H} \rrbracket^w(\{p\}) \\ &= \lambda x_e. \lambda w: \underline{p \in C}. \left[\begin{array}{c} p \in C \wedge \\ \mu_w^{\text{hope}}(x, p) > \theta^{\text{hope}}(C) \end{array} \right] \end{aligned}$$

The resulting truth conditions presuppose and assert that the prejacent *p* of the polar interrogative is part of the comparison class *C*, and assert furthermore that *x*'s degree of preference for *p* exceeds a standard threshold for the comparison class, provided by θ^{hope} . If we had instead canonically composed *hope* with the ordinary content of a polar interrogative, we would have obtained the trivial truth conditions reviewed in Section 2.1. If we assume that this non-canonical composition strategy is a last resort strategy that is made available by the need to avoid ungrammaticality as caused by these trivial truth-conditions, and that this availability is subject to variation, this analysis of *hope whether p* sentences can account for their Marginal/Variable Acceptability. Furthermore, this analysis also straightforwardly accounts for the Interpretive Asymmetry of *hope whether p*, because it is predicted to be truth-conditionally equivalent to *hope that p*.¹⁵

¹⁵Under the assumption that constituent questions do not highlight a single proposition but rather an individual property (Roelofsen and Farkas, 2015; Theiler, 2021), the analysis formulated above predicts that *hope* cannot take constituent questions due to a type mismatch. However, in principle one can reformulate the lexical entry of *hope* to allow it to compose with a highlighted content that is an individual property (e.g., by first existentially quantifying over it, cf. Theiler, 2021). Therefore, a highlighting-based analysis need not predict incompatibility between *hope* and constituent questions (and this point similarly applies to alternative questions).

Empirically, such examples are attested (i) and their acceptability also varies among speakers. In this paper, we focus on

However, despite its initial appeal, this analysis has two limitations that make it less preferable. First, since *hope whether p* is predicted to be truth-conditionally equivalent to *hope that p*, the analysis does not immediately account for the Inquisitive Implication (58c) of the former. One may hope to derive the Inquisitive Implication of *hope whether p* from its pragmatic competition with *hope that p*, but spelling out the details and properly motivating the necessary assumptions can turn out to be non-trivial. For instance, it might be tempting to derive the Inquisitive Implication of *hope whether p* by appealing to Levinson’s (2000) M-principle. However, while *hope whether p* is indeed marked when compared with *hope that p*, it is not obvious that the attitude holder wondering whether *p* is the abnormal/non-stereotypical situation when they hope that *p*.

Second, as we will see in the next section, there are other constructions that similarly exhibit Interpretive Asymmetry. However, for such constructions, it is implausible to derive Interpretive Asymmetry by stipulating non-canonical composition with the highlighted content of the polar question. That is, the analysis presented here cannot generalize to those constructions. In contrast, the alternative analysis we will present in the next section can account for the Interpretive Asymmetry for those constructions as well as the *hope whether* sentences here. Therefore, the alternative analysis is more parsimonious and should be preferred.

4.3 Candidate analysis 2: adjunction

4.3.1 Turkish *diye* and Japanese *to* interrogative clauses

The starting point of our second candidate analysis concerns data from Turkish and Japanese. In these two languages, there are clauses introduced by the morphemes *diye* (Turkish) and *to* (Japanese), which very naturally allow for *hope* to combine with interrogatives, as seen in (71). As far as we can tell, the Turkish and Japanese data are parallel on the points relevant here, so we only illustrate with the former.

- (71) a. John [komşu-su ev-de ol-acak mı diye] um-uyor
 John neighbor-3S.POSS house-LOC be-FUT.3S Q DIYE hope-IPFV.PRES.3S
 “John hopes whether his neighbor will be home.”
 b. [Bana birileri yardımcı ol-abil-ir mi diye] um-uyor-du-m.
 1S.DAT someone helpful be-MOD-AOR.3S POLQ DIYE hope-IPFV-PST-1S
 “I was hoping whether someone could help me.”
 c. γ[O sebeple Hitman’de de ol-ur mu diye] um-du-m ama hayal kırıklığı
 that’s why Hitman.LOC too be-AOR POLQ DIYE hope-PFV.PST-1S but disappointment
 yaşa-dı-m.
 live-PFV.PST-1S
 Lit. “That’s why I hoped whether it [=a Turkish translation] would be available in Hitman too,
 but I was disappointed.” (Turkish)

Examples (71a) and (71b) contrast with the ones in (72), where the interrogatives are realized by nominalized clauses composing with the predicate *um-*.

- (72) a. *[Bana birileri-nin yardımcı ol-up ol-ama-yacağ-ın-ı] um-uyor-du-m.
 1S.DAT someone-GEN helpful be-COORD be-NEG.MOD-NMZ.FUT-3S-ACC hope-IPFV-PST-1S
 Intended: “I was hoping whether someone could help me.”
 b. *[O sebeple Hitman’de de ol-up ol-ma-yacağ-ın-ı] um-du-m.
 that’s why Hitman.LOC too be-COORD be-NEG-NMZ.FUT-3S-ACC hope-PFV.PST-1S
 Intended: “That’s why I hoped whether it would be available in Hitman too.” (Turkish)

The first step in Özyıldız and Uegaki’s (2023) analysis of such clauses is the proposal that nominalized clauses *hope whether* examples and leave other types of interrogative clauses under *hope* for future research.

- (i) a. γBusinesses that sell to the U.S are wondering and hoping when the U.S. economy will get back into the game.
 (https://toronto.citynews.ca/2020/08/11/business-highlights-450/)
 b. γSo here I sit two weeks into my recovery wondering and hoping when the next time I’ll be standing on top of a
 Scottish Mountain will be! (https://ravensgully.com/2014/11/10/pesky-peroneals/)

are arguments of the attitude predicates that they combine with, whereas *diye/to*-clauses are modifiers. One of the main empirical motivations for this difference comes from the observation that *diye*-clauses may combine with (lexically or derived) intransitive predicates like ‘walk around,’ in (73a), whereas this is not an option for nominalized clauses, as shown by the unacceptability of (73b). (Compare this to the acceptable examples of nominalized clauses combining with attitude verbs seen in Section 3.)

- (73) a. [Bana birileri yardımcı ol-abil-ir mi diye] dolan-dı-m.
 1S.DAT someone helpful BE-MOD-AOR.3S POLQ DIYE walk around-PFV.PST-1S
 “I walked around wondering whether someone could help me.”
 b. *[Bana birileri-nin yardımcı ol-up ol-ama-yacağ-ın-ı]
 1S.DAT someone-GEN helpful be-COORD be-NEG.MOD-NMZ.FUT-3S-ACC
 dolan-dı-m.
 walk around-IPFV-PST-1S
 Intended: “I walked around wondering whether someone could help me.” (Turkish)

We note that at least Washo may display a similar pattern in its clause-type inventory (Bochnak and Hanink, 2022) and that the facts that we report on here are also discussed by Yıldırım-Gündoğdu (2018) for Turkish, Saito (2012, 2015), Shimamura (2018), Kim (2018), Tomioka and Kim (2020, 2024) and Goodhue and Shimoyama (2022) for Japanese.

The grammaticality of (73a) suggests that a *diye*-clause can be integrated with any verb phrase as an adjunct, including the ones in (71). Importantly, the *wondering* in the translation of (73a) does not correspond to any lexical predicate in the Turkish string, and its meaning is argued to be contributed by the morpheme *diye* together with its question complement. We follow Özyıldız & Uegaki in assuming that *diye* generally contributes a linguistic production inference, even with predicates like *um-* (‘hope’), and model this inference with the predicate $\lambda e.utter(e)$ below. (Linguistic production is understood here in a very broad sense, and includes speech, thoughts, signs, etc.)

For the truth conditions of examples like (73a) and (71b), we simplify Özyıldız and Uegaki’s (2023) proposal into (74).

- (74) a. [I [[can someone help me? *diye*] [walk around]]]
 Lit. I walked around wondering whether (*diye*) someone could help me. = (73a)
 $\exists e, e_1, e_2 : e = e_1 + e_2 \wedge walk-around(e_1) \wedge utter(e_2) \wedge$
 $content(e_2) = \llbracket \text{Will someone help me?} \rrbracket \wedge author(e) = Sp$
 b. [I [[can someone help me? *diye*] [P hope]]]
 I was hoping whether (*diye*) someone could help me. = (71b)
 $\exists e, e_1, e_2 : e = e_1 + e_2 \wedge hope(e_1) \wedge utter(e_2) \wedge$
 $content(e_1) = P \wedge content(e_2) = \llbracket \text{Will someone help me?} \rrbracket \wedge author(e) = Sp$

Thus, attitude reports with *diye* clauses involve the summation of two events: One contributed by a main verb (*walk around* or *hope*) and one contributed by the morpheme *diye* (*utter*, here). The content of the latter is determined by the clause that *diye* introduces and the subject of the summed event is the same. Note that we understand the predicate *um-* (‘hope’) to be transitive here, and provide it with a silent pronoun *P* serving as its internal argument. This pronoun is left unbound, and is understood to be valued by context.

Inspired by the above analysis of *hope diye/to whether* sentences in Turkish and Japanese, our second analysis of English *hope whether* sentences (and their Mandarin Chinese counterparts) assumes that such sentences involve non-canonical composition of *hope* and the polar interrogative yielding truth conditions along the lines of (74b). In effect, the polar interrogative is understood as a speech/thought report, which is further integrated with the predicate *hope* as an adjunct. Note that we need not assume that those English speakers who accept *hope whether* sentences necessarily have a silent counterpart of *diye/to* as part of their grammars. For our purposes, we only assume that (with enough contextual support) they are able to obtain (74b) as the non-canonical/coerced interpretation of *I was hoping whether someone could help me*, which avoids the trivial interpretation derived by canonical composition. Assuming that this is only a last resort and its availability is subject to variation, the analysis can account for the Marginal/Variable Acceptability

of *hope whether* sentences.¹⁶

Furthermore, given that in (74b) linguistic production can be understood as the expression of thoughts, this analysis can straightforwardly account for the Inquisitive Implication (58c) of *hope whether* sentences. The remaining empirical property to be accounted for is Interpretive Asymmetry, which we discuss in the next two sections.

4.3.2 Interpretive Asymmetry: need for a pragmatic account

In this section, we show that Interpretive Asymmetry holds for two constructions that involve evaluatively positive predicates in addition to English *hope whether p* and its Mandarin counterpart. Based on this data, we argue that a pragmatic account for the asymmetry is needed.

We start with the Turkish *diye*-examples in (71). The crucial observation is that they only allow interpretations where the relevant hope targets the positive answer to the question. To see this more closely, consider example (75a) (= (71a)) in the GOODFRIEND and NOISEHATER contexts we introduced in Section 4.1. The example is judged true in the former context, but false in the latter, suggesting that it may only express John’s hope for his neighbor to be home.

- (75) a. John [komşusu ev-de ol-acak mı diye] um-uyor.
 John his neighbor house-LOC be-FUT.3S Q DIYE hope-IPFV.PRES.3S
 “John hopes whether his neighbor will be home.” (Turkish)
- b. Truth value judgment in...
 (i) the GOODFRIEND context: True;
 (ii) the NOISEHATER context: False.

Diye-clauses may also combine with the predicate *kork-*, “fear,” as shown in (76). In contrast with the one in (75), this kind of sentence does not exhibit Interpretive Asymmetry: Example (76) is judged true in the GOODFRIEND context in which John wants his neighbor to be home (*fear that not p*), as well as the NOISEHATER context in which he doesn’t want his neighbor to be home (*fear that p*).

- (76) a. John [komşusu evde olacak mı diye] korkuyor.
 John his neighbor at home be-FUT Q DIYE fears
 “John fears whether his neighbor will be home.” (Turkish)
- b. Truth value judgment in...
 (i) the GOODFRIEND context: True;
 (ii) the NOISEHATER context: True.

Turning to our second construction, the same contrast between evaluatively positive and negative predicates regarding Interpretive Asymmetry is observed in the interpretation of adverbs like *hopefully* and *fearfully* in English when they modify the predicate *wonder*. For instance, consider the following context (77), according to which Des prefers John to be dead while Ali prefers John to be alive.

- (77) Des, an assassin, sabotages the brakes of the vehicle transporting John, the star witness in a criminal case, and watches it careen off a cliff. Shortly after, Ali, the prosecutor in the case, arrives with a rescue team. Des hadn’t expected Ali to arrive so quickly. While it’s clear that John is seriously injured, Des isn’t sure whether he’s dead. If John survives and is saved by the rescue team, Des knows he will face severe consequences for failing the assassination attempt.

Let $?p$ be a polar question with radical p . We observe that x *wonders hopefully* $?p$ is felicitous only if the attitude holder x hopes that p : (78a) is incompatible with context (77) because it implies that Des hopes that John is alive, and similarly (78d) is infelicitous because it implies that Ali hopes that John is dead.

- (78) a. #Des wonders hopefully: “Is John alive?” ($\rightsquigarrow$ Des hopes that John is alive.)
 b. Des wonders hopefully: “Is John dead?” ($\rightsquigarrow$ Des hopes that John is dead.)
 c. Ali wonders hopefully: “Is John alive?” ($\rightsquigarrow$ Ali hopes that John is alive.)

¹⁶Crucially, Turkish and Japanese are different from English and Mandarin Chinese in that the adjunction-style composition is canonical for *diye/to*-clauses.

- d. #Ali wonders hopefully: “Is John dead?” ($\rightsquigarrow$ Ali hopes that John is dead.)

In contrast, x *wonders fearfully* $?p$ is compatible with the attitude holder having a fear towards p as well as towards $\neg p$. All the sentences below are compatible with the context. Note that for (79b) and (79c), where the attitude holder’s fear is towards the negation of the radical, some consultants initially found them slightly weird but ultimately judged them true and felicitous in the context. Therefore, even though (79b) and (79c) may not be ideal and may initially appear to be infelicitous, they are still compatible with (77). They are crucially different from (78a) and (78d), which are clearly infelicitous.

- (79) a. Des wonders fearfully: “Is John alive?” ($\rightsquigarrow$ Des fears that John is alive.)
 b. (?) Des wonders fearfully: “Is John dead?” ($\rightsquigarrow$ Des fears that John is not dead.)
 c. (?) Ali wonders fearfully: “Is John alive?” ($\rightsquigarrow$ Ali fears that John is not alive.)
 d. Ali wonders fearfully: “Is John dead?” ($\rightsquigarrow$ Ali fears that John is dead.)

We have seen that Turkish *hope diye whether* p and English *wonder hopefully* $?p$ both exhibit Interpretive Asymmetry. What they have in common is that the interrogative clause is not integrated as the internal argument of *hope(fully)*. That is, there is no direct compositional semantic relation between the preference described by *hope(fully)* and the interrogative clause. Therefore, it would not work to stipulate a semantics of *hope(fully)* that directly takes the highlighted content of an interrogative clause. Rather, its sensitivity to the highlighted content should be achieved indirectly, based on independently motivated pragmatic principles regarding how a speaker should choose the radical/highlighted content of a polar question.

Since (80a) and (80b) each (contextually) entail (81), we only need to account for why (81) is only compatible with Ali hoping that John is alive (and the analysis should also correctly capture the difference between *wonder hopefully* and *wonder fearfully*). In the next section, we present such a pragmatic analysis.

- (80) a. Ali [John hayatta mı diye] umdu.
 Ali John alive Q DIYE hoped
 Lit. Ali hoped (*diye*) whether John was alive.
 b. Ali is hoping whether John is alive.
 c. $\exists e, e_1, e_2 : e = e_1 + e_2 \wedge \text{hope}(e_1) \wedge \text{utter}(e_2) \wedge$
 $\text{content}(e_1) = P \wedge \text{content}(e_2) = \llbracket \text{Is John alive?} \rrbracket \wedge \text{author}(e) = \text{Ali}$
- (81) Ali wonders hopefully: “Is John alive?”
 $\exists e [\text{agent}(e) = \text{Ali} \wedge \text{wonder}(e) \wedge \text{hopeful}(e) \wedge \text{content}(e) = \llbracket \text{Is John alive?} \rrbracket]$

4.3.3 A pragmatic analysis of interpretive asymmetry

In this section, we present a pragmatic account of why x *wonders hopefully* $?p$ is only compatible with x hoping that p , as well as why there is no such a constraint for *wonder fearfully*.

For concreteness, we adopt Tabatowski’s (2022) analysis of canonical polar questions, i.e., those without syntactically high negations. The analysis informally says that asking *whether* p expresses the speaker x ’s desire to learn p if p is true. Formally, a polar question with radical p has the following expressive content.

- (82) $\llbracket ?p \rrbracket_x^{w,t} = \forall w' \in \text{Dox}_x^{w,t} [p(w') \rightarrow (\text{Sim}^{w'}(\lambda v. \text{ctb}^{v,t}(p)(x)) <_{\text{Goals}^w(x)} \text{Sim}^{w'}(\lambda v. \neg \text{ctb}^{v,t}(p)(x)))],$
 where $\text{ctb}^{v,t}(p)(x)$ is true iff x comes to believe p in v at t .

According to (82), when x asks whether p , they express that in every p -world w' in their doxastic state, the set of minimally different worlds where x comes to believe p is better (with respect to x ’s goals) than the set of the minimally different worlds where x does not come to believe p .¹⁷ Here the ordering over worlds

¹⁷There are two main differences between (82) and Tabatowski’s (2022) original fomulation below (his ex. 21 on p. 15).

- (i) $\llbracket ?p \rrbracket_x^w = \forall w' \in \text{Dox}_x^w [p](\text{Sim}^{w'}(v)(\text{ctb}^{v,t}(p)(x)) <_{\text{Goals}^w(x)} \text{Sim}^{w'}(v)(\neg \text{ctb}^{v,t}(p)(x)))$

First, we use $\text{Sim}^{w'}(\lambda v. \text{ctb}^{v,t}(p)(x))$ to maintain the more familiar notation of a similarity selection function where $\text{Sim}^{w'}(q)$ returns the set of q -worlds closest to w' . Second, Tabatowski uses the notation $\text{Dox}_x^w[p]$ to represent p -worlds in x ’s doxastic state, whereas we maintain the standard notation of Dox_x^w and achieves the intended effect of restricting the quantificational domain to p -worlds by adding a material implication in the scope.

$<_{\text{Goals}^w(x)}$, which can then be straightforwardly lifted to compare sets of worlds, is standardly defined using $\text{Goals}(x)$ as the ordering source: $w_1 <_{\text{Goals}^w(x)} w_2$ iff more goals in $\text{Goals}^w(x)$ are satisfied in w_1 than in w_2 . For brevity, we will simply say that the agent takes worlds where they (correctly) learn p to be closer to achieving their goals than worlds where they do not learn p , even though strictly speaking the comparison is only among worlds that are minimally different from the p -worlds in their doxastic state.

This analysis is motivated by and used to account for, among other things, Bolinger’s (1978) extended paradigm, whereby a polar question with radical p can be used as an offer (83) of or a request (84) for p but not $\neg p$ (van Rooij and Šafářová, 2003; AnderBois, 2011; Tabatowski, 2022):

- (83) Offer of p :
 - a. Do you like beer?
 - b. #Do you not like beer?
- (84) Request for p :
 - a. Do you have sparkling water?
 - b. #Do you not have sparkling water?

As an example, we show how Tabatowski’s analysis derives the contrast in (83). In this context, one can reasonably assume that the speaker’s (ultimate) goal is to serve the addressee something they like. Call this goal G_0 .

Let us start with (83a). Tabatowski assumes that the ordering source $\text{Goals}(x)$ not only contains the speaker’s ultimate goal(s) but also their subgoals (to be defined shortly). To see that G_0 alone is not enough to satisfy the requirement (82), imagine that there are two types of drinks available: beer and sparkling water, and that the addressee may like beer or water or both or neither. Then, consider a world w' in the speaker’s doxastic state, where the addressee likes both beer and water and the speaker will serve them (only) water without knowing anything about what the addressee likes.¹⁸ With respect to G_0 alone, the closest worlds to w' where the speaker learns that the addressee likes beer are just as good as (and thus not better than) those where the speaker does not learn so: either way the addressee will get water, which is something they like, and thus G_0 is met. Therefore the requirement (82) is not met if the ordering is based on G_0 alone, but (83a) is clearly a felicitous offer. In light of this, Tabatowski proposes that subgoals, which he defines as what the agent takes to be necessary to achieve their ultimate goal, are also used to induce the ordering over worlds. When the ultimate goal is G_0 , Tabatowski assumes that a plausible subgoal G'_0 is for the speaker to know something that the addressee likes.¹⁹ When this subgoal G'_0 is included, the speaker learning that the addressee likes beer is indeed better than not learning so (precisely because the former meets this subgoal but the latter does not), and therefore the requirement (82) is met. This accounts for the felicity of (83a).

Now we turn to (83b). Given the same contextual goal G_0 and its subgoal G'_0 , the speaker learning that the addressee does not like beer is no better than not learning so (since G_0 will have the same truth value in both cases, and G'_0 is not met in either case). Therefore, the requirement (82) is not met and hence (83b) is infelicitous as an offer.²⁰ Note that this does not mean that (83b) is generally infelicitous. The analysis above only says that it cannot be used with the goal being G_0 . But there can be other goals that satisfy

¹⁸The speaker’s lack of knowledge about what the addressee likes in w' is a consequence of *negative introspection*, i.e., $\forall \varphi. (\neg B\varphi \rightarrow B\neg B\varphi)$. When (83a) is felicitously uttered in w , the speaker is ignorant about what the addressee likes. Then, given negative introspection, the speaker is also ignorant about what the addressee likes in any world w' in their doxastic state at world w .

¹⁹Tabatowski does not spell out in what sense the subgoal G'_0 is a necessity condition for G_0 . Since he assumes that w' , where G'_0 is not met, is in the speaker’s doxastic state, G'_0 is clearly not a doxastic necessity. We suggest that the relevant notion is that G'_0 is necessary to *guarantee* the achievement of G_0 . Even though the speaker does not rule out the doxastic possibility that they will randomly serve the addressee something without knowing what they like and it just so happens that they like it, this is not a reliable way to achieve their ultimate goal. In order to guarantee the achievement of their ultimate goal (i.e., to make it an *action goal* in the sense of Alonso-Ovalle and Menéndez-Benito 2018), in this case the speaker would need to know something that the addressee likes so that they can serve the addressee accordingly.

²⁰Crucially, the assumption that subgoals are necessary conditions for (guaranteeing) goals prevents them from being too fine-grained. Suppose one were to include finer-grained subgoals of the form *knowing whether the addressee likes X*, where X ranges over types of beverages. The requirement (82) would be met even for (83b), because learning that the addressee does not like beer achieves one of the subgoals whereas not learning it achieves none. However, knowing whether the addressee likes beer is not necessary to guarantee G_0 , because it may be that the speaker knows that the addressee likes water without knowing whether they like beer. Therefore knowing whether the addressee likes beer will not in fact be included as a subgoal.

the requirement (82). Most notably, the speaker may have a purely epistemic goal of learning whether the radical is true ($K?p$).²¹ The requirement (82) is indeed met for (83b) with respect to $K?p$: when the radical is true, i.e., when the addressee does not like beer, it is indeed better for the speaker to learn this (than for them not to learn this) because the goal $K?p$ is met. But crucially, in this case the question (83b) is not construed as an offer.

Having shown Tabatowski’s analysis for matrix polar questions, below we show how it can be extended to embedded cases to account for the contrast between *hopefully* and *fearfully*. For our purposes, Tabatowski’s analysis can be incorporated in an event semantic framework (85), according to which (82) must hold for an event of wondering $?p$ (assuming the agent follows this convention on polar questions).

$$(85) \quad \forall e [(agent(e) = x \wedge wonder(e) \wedge content(e) = \llbracket ?p \rrbracket) \\ \Rightarrow \forall w' \in \text{Dox}_x^{w,t} [p(w') \rightarrow (\text{Sim}^{w'}(\lambda v. \mathbf{ctb}^{v,t}(p)(x)) <_{\text{Goals}(e)} \text{Sim}^{w'}(\lambda v. \neg \mathbf{ctb}^{v,t}(p)(x))]]$$

Assuming the following semantics (86) for *wonder hopefully/fearfully* $?p$, we will show that while (86b) is compatible with (85) no matter whether the agent fears p or $\neg p$, (86a) is compatible with (85) only when the agent hopes p .

$$(86) \quad \begin{array}{ll} \text{a.} & \exists e [agent(e) = x \wedge wonder(e) \wedge content(e) = \llbracket ?p \rrbracket \wedge hopeful(e)] \\ \text{b.} & \exists e [agent(e) = x \wedge wonder(e) \wedge content(e) = \llbracket ?p \rrbracket \wedge fearful(e)] \end{array}$$

In a nutshell, our analysis relies on a key difference between hopeful and fearful events: For (86b), fearful events are not characterized by goals, but by *threats*. In particular, *fearfully* does not identify the goals of the wondering event, and this allows the ordering source in (85) to consist of purely epistemic (not bouletic) goals. In contrast, for (86a), *hopefully* necessarily attributes to the agent a set of bouletic goals that are shared by the wondering event. These constrain the ordering source in (85) and thus constrain the interpretation.

We start with *fearfully*. We first argue that *fear*, unlike *hope*, is not characterized by goals. Suppose one of x ’s goals is to win a prize in a lottery, where winning nothing is likely. In this case, it is true that x hopes they will win a prize, which is equivalent to x hoping that they will not win nothing. But it is not necessarily true that x fears they will win nothing: this depends on whether x finds winning nothing a prospect undesirable enough to be considered fear-inducing (which we may call a *threat*).

Consequently, we expect that *fearfully* does not identify the goal(s) of the event it modifies, and this is indeed borne out. In (87a), Mary’s knocking need not have a goal of avoiding her fear: it may well be that she fears what her knocking may bring about. In contrast, *hopefully* does identify the goal(s) of the event it modifies (87b), which we will further discuss later.

$$(87) \quad \begin{array}{ll} \text{a.} & \text{Mary knocked fearfully on the door.} \not\approx \text{She knocked in order to avoid what she feared.} \\ \text{b.} & \text{Mary knocked hopefully on the door.} \approx \text{She knocked in order to obtain what she hoped.} \end{array}$$

Given that fearful events are characterized not by goals but rather by threats, in (86b) the witness event e being a fearful one only requires that the goals of e as a wondering event be compatible with the threats. This means that e , being a wondering event, can have a pure epistemic goal $K?p$ (i.e., to know whether the radical p is true). In this case, we can verify that (86b) is indeed compatible with (85): p -worlds where the agent comes to believe p are indeed better than those where they do not, because $K?p$ is met in the former but not the latter. Crucially, this holds regardless of whether x fears the radical p or its negation. Therefore all sentences in (79) are technically compatible with the context (77), as long as the wondering event is also a fearful event.²²

Now we turn to *hopefully*. In (87b), Mary’s knocking is interpreted as being in service of her goal(s). More concretely, we observe that the goal may be what she hopes for, e.g., in (88a) she hopes to be let in and the goal of her knocking is to be let in, or for her to learn that what she hopes for is true, e.g., in (88b) Mary hopes that Bob is his room and the goal of her knocking is for her to learn that this is indeed the case.

²¹It is also possible that the speaker’s goal is to serve the addressee something that they *don’t* like, as claimed by Tabatowski (2022, p. 9, fn. 2).

²²Regarding the possible preference for (79a) and (79d) for some speakers, given that the radical in a polar question is made particularly salient (Roelofsen and Farkas, 2015; Krifka, 2013), and hence presumably easier to entertain than its negation, it would be more straightforward to construe the wondering event as a fearful one if the radical is what the agent fears.

- (88) a. Context 1 for (87b): Mary just returns home and realizes that she has lost her key. However, she knows that her roommate Bob is usually home at this time.
 b. Context 2 for (87b): Mary just learns that there was an explosion at the gas station where her roommate Bob works. She remembers that Bob told her that he would take a sick leave today and stay in bed.

Based on this, we propose the descriptive generalization (89) that constrains the goals associated with the event *hopeful* modifies.²³

- (89) In a hopeful event e where the agent hopes φ , the goal of the event is either φ or $B_{agent(e)}\varphi$.

Now we can derive the constraint on interpretation for *wonder hopefully* ? p . We first show that (86a) is compatible with (85) when x hopes p . From (89) we know that the agent can have Bp as a goal. This would meet the requirement (85) because p -worlds where the agent learns p is indeed better than those where they don't (since Bp is met in the former but not the latter). We then show that, in contrast, (86a) is incompatible with (85) when x hopes $\neg p$. In this case (89) instead tells us that, in light of the constraint imposed by the event being hopeful, there are two potential goals for the wondering event: $\neg p$ and $B\neg p$. However, regardless of which goal(s) we consider, p -worlds where the agent learns p is no better than those where they don't, because in both cases neither goal is met. Therefore the requirement (85) is not satisfied. This accounts for the Interpretive Asymmetry for *wonder hopefully* ? p .

5 General discussion and conclusion

In describing the distribution and the interpretation of embedded interrogatives with different preferential predicates, we find support for the view that the restrictions and freedom in the combinatorial properties of clause-embedding predicates are mostly, if not entirely, governed by their semantic properties. While this is a point of consensus in the semantic literature on clause-embedding, disagreements remain about which semantic properties are relevant (veridicality? preferentiality? neg-raising? eventivity? and so on), and about which generalizations are empirically motivated that relate clause-embedding predicates' semantic and combinatorial properties.

One point of tension is between Uegaki and Sudo (2019), who suggest that all non-veridical preferential predicates have to be anti-rogative, and White (2021), who produces examples where *hope* and *fear*, two paradigmatic non-veridical preferential predicates, are attested with embedded questions. This suggests that the original empirical generalization is false and leads to the natural conclusion that its explanation might be misguided, and that we should look for properties other than (non-)veridicality and preferentiality to explain the combinatorial properties of attitude predicates.

In this paper, we take this challenge seriously. In addition to spelling out the interpretive properties of *hope* and *fear* when they combine with embedded questions, we expand the repertoire of counter-examples to Uegaki and Sudo's (2019) empirical generalization and check the behavior of these predicates' counterparts in Japanese, Mandarin, Spanish and Turkish. Concretely, we propose an updated classification of NVPs based on two lexical semantic properties: C(lausal)-distributivity and valence, that is, whether the equivalence holds that $\lceil x Vs Q \rceil$ iff there is an answer p to Q s.t. $\lceil x Vs that p \rceil$, and whether a preferential predicate V is evaluatively positive or negative. Predicates that are either non-C-distributive or negative are generally free to combine with any form of embedded questions. We consider these to be genuine counter-examples to Uegaki and Sudo's (2019) empirical generalization but point out that Uegaki and Sudo's formal analysis in fact makes the right predictions about these predicates, because the relevant semantic properties that would lead to triviality when NVPs take questions in fact do not hold for such predicates. Among such responsive NVPs, the C-distributive ones, which are evaluatively negative, describe a relation between individuals and any possible answer to an interrogative (i.e., *fear whether p* may either be paraphrased as *fear that p* or *fear that not p*), and the non-C-distributive ones, a relation between individuals and questions that is not reducible to relation to any particular answer. The remaining class of NVPs, i.e., those that are both positive and C-distributive such as *hope*, are difficult to combine with embedded questions. This leads us to maintain

²³This generalization may be further derivable from the close connection between hopes and goals, but we leave a detailed analysis for future research.

Uegaki and Sudo’s (2019) original proposal for these predicates, which appeals to the Threshold Significance Presupposition to derive that they should be anti-rogative (see Section 2.1), and to argue that when we do observe attested examples where *hope* appears to embed questions, the question is not composed with the predicate in a canonical way, i.e., as its semantic argument. Rather, a non-canonical composition mechanism is invoked as a last resort, where the question is interpreted as a speech/thought report and is integrated with the predicate as an adjunct. Such an analysis is independently motivated by the Japanese and Turkish data, where attitude predicates can combine with clauses whose meanings are enriched with the inclusion of a speech event. We also demonstrate how the asymmetric interpretation of such examples can be derived once we take into account the pragmatics of asking questions.

To corroborate or call into question our current findings, it is crucial to expand our search space to more predicates, and to languages different from the ones surveyed here. If our findings here are on the right track, the prediction is that predicates that trigger the Threshold Significance Presupposition and that are C-distributive will be restricted in their ability to combine with embedded questions—in line with Uegaki and Sudo’s (2019) original observation. (We have only found evaluatively positive predicates with the Threshold Significance Presupposition, but do not exclude the possibility that there might be negative ones as well.) In contrast, predicates that lack either one of these properties are expected to be freer in their combinatorial properties.²⁴

Equally important is the notion that languages might offer different strategies for combining attitude predicates with clauses. We have argued that these differences may have to do with whether a given strategy is part of the core grammar of a language or whether it is one of last resort, and whether combination proceeds through the basic lexical denotation of attitude predicates, or through an enriched one (e.g., enriched with the conjunction of a speech report). We leave a more refined investigation of (the effects of) these claims for further research.

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²⁴But there may well be additional syntactic and semantic factors that make them incompatible with questions. Also, it is important to keep in mind that one should not take for granted that a predicate *P* in one language and its intuitive counterpart *P'* in another language will share the relevant semantic properties discussed in this paper. For instance, if a language only has a non-C-distributive positive predicate similar to Mandarin *qidai* or Japanese *tanosimi* (i.e., it does not have a predicate similar to *hope*), speakers of that language may use such a predicate to translate *hope*. In this respect, semantic properties such as C-distributivity can help diagnose whether a predicate’s intuitive counterpart in a different language is really equivalent.

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A Non-veridical *care*

Elliott et al.’s (2017) argument shows that *emotive factive* uses of *care* are non-C-distributive. We further suggest that English *care* also has uses that are not emotive factive. For simplicity, we will call such uses *non-veridical*. Consider the following naturally-occurring example where a law firm partner talks about qualities they look for.

- (90) γ “[...] I want people who will do what we ask them to do — not people who feel they have to be mentored and rewarded all the time, and who need to be patted on the head every time they do a brief.”
 “I don’t care if you got an A in con law,” she continued. **“I care that you will work hard.”**

Here, *care* is non-veridical in *I care that you will work hard*: It does not entail that the addressee will work hard or that the attitude holder (which is the speaker in this case) believes that the addressee will work hard.²⁵ The attitude holder merely wants it to be the case that the addressee will work hard.

One crucial difference between veridical and non-veridical uses of *care* is whether the attitude holder must have a preference for the embedded clause to be true. For veridical uses of *care*, it is possible that the attitude holder does not prefer that the embedded clause be true (91).

- (91) Al cares that Bo left (without notifying him), which is why he is so mad at Bo.

In contrast, we observe that non-veridical uses of *care* entail that the attitude holder wants the declarative complement to be true (92). For instance, in a non-veridical context (93), given that Al (not knowing whether Bo left) wants it to be the case that Bo left, (93a) is judged acceptable. This is because (93a) is true under a non-veridical use of *care*. In contrast, (93b) is not acceptable in this context, because it is not true under the non-veridical use.

- (92) Al cares that *x* left
 $\Rightarrow$ Al wants it to be the case that *x* left (under non-veridical use)
- (93) Non-veridical context: Al really dislikes Bo and wants to avoid him as much as possible. Al is invited to a party. He knows that Bo is also invited and that Bo generally comes to a party early and does not stay for long. Therefore Al decides to come to the party after Bo has left. After waiting for some time, Al calls his friend Charles, who is at the party, to check whether Bo is still there. When Charles (not knowing Al’s beef with Bo) picks up the phone, he says: “Why aren’t you here already? The beer is amazing!”
 Al replies: “I don’t care if they serve the best beer in the world there. **I care that Bo has left.** (Has he left or not?)”
 a. $\checkmark$ Al cares that Bo has left. (Non-veridical use: true)
 b. $\#$ Al cares that Bo is still there. (Non-veridical use: not true)

²⁵Technically, in the original context (90), the speaker is using *you* in a generic way. But the sentence would also be felicitous if the speaker is talking to a new employee and intends *you* to specifically refer to the addressee.

Based on the observation that non-veridical uses of *care* entail the attitude holder wants the declarative complement to be true, we can show such uses are also non-C-distributive: (94) can be true without there being a student x such that Al wants it to be the case that x left. For instance, (94) is true in context (95), whereas (96) is not true in this context for any student x .

- (94) Al cares which student left.
- (95) Context: Al's job is to record which student left and he definitely wants to do it right. However, Al does not have any preference about which student left. That is, it is not the case that there exists a student x s.t. Al prefers it to be true that x left.
- (96) Al cares that x left.
 $\Rightarrow$ Al prefers it to be the case that x left. (under non-veridical use of *care*)

The discussion above suggests that non-veridical uses of *care* are evaluatively positive, and if a question-taking *care*, such as the one in (94), could have a non-veridical use, it would be non-C-distributive. However, given that in most cases the salient reading of *care* is the veridical one, one may still question whether the same *care* is involved in its non-veridical uses and when it takes questions in (94). Therefore, we only take non-veridical uses of *care* as a tentative example of an evaluatively positive non-C-distributive NVP.

B References to naturally occurring examples

- Exx. (18) & (19): https://www.ximalaya.com/ask/q8788545?source=m_jump
- Ex. (20a): <https://www.smbc.co.jp/sponsorship/smbcniipponseries/column/article01/>
- Ex. (21a): https://www.marvelvsdc.xyz/entry/dcfu_89
- Ex. (26a): <https://www.nytimes.com/1972/09/08/archives/painted-furniture-new-interest-in-an-old-art.html>
- Ex. (26b): <https://www.thefreelibrary.com/Our+pick+of+the+best+soaps+%26+telly.-a0295046793>
- Ex. (26c): <https://www.city-data.com/forum/las-vegas/340170-rainbow-family-moving-vegas-30-days-2.html>
- Ex. (26d): <https://tvmegasite.net/updates/days/2004/days-12-10-04.shtml>
- Ex. (43a): <https://eu.heraldtribune.com/story/news/2012/11/11/on-scats-express-bus-it-s-life-in-the-fast-lane/29134294007/>
- Ex. (71c): <https://www.technopat.net/sosyal/konu/hitman-2-neyin-devami.1236026/post-8635696>
- Ex. (90): <https://www.lawpracticetoday.org/article/right-kind-grit-succeed/>